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Review

Charging station location problem: A comprehensive review on models and solution approaches

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ABSTRACT

Charging infrastructure planning has a strategic impact on promoting the use of electric vehicles (EVs) and other alternative fuel vehicles. Importantly, decision makers need to answer the question on the number and location of charging stations in a way to satisfy customer recharging demand and meet certain restrictions imposed by real-life considerations. In this context, we consider the charging station location problem (CSLP), which belongs to the category of facility location problems and seeks to optimize the locations of charging stations. A growing body of literature has developed on this subject in recent years. Various approaches have been proposed to model the problem taking into account different features, constraints, decisions and performance measures as well as the dynamic and stochastic components inherent to the problem. Moreover, efficiently solving CSLP, particularly when applied to real-life case-studies, might be challenging in practice. Considerable effort has thus been made by researchers to develop innovative solution methods based on exact or heuristic approaches to obtain good quality solutions within short computation times. Therefore, we provide in this paper a comprehensive review on the literature relevant to CSLP, with a particular focus on modeling and solving the problem. We analyze the literature from different perspectives including demand representation, demand coverage approaches, objective functions, side constraints, decision variables, model structure as well as time dependency and uncertainty on the problem parameters. We also present various ways of classifying existing works, which allows readers to capture different aspects of the problem that researchers have tended to focus on and identify opportunities for further developments. We believe our work could be helpful to researchers by providing an overview of the CSLP literature and suggesting perspectives for future research in the field.

1. Introduction

1.1. Motivation and background

As transportation activity is responsible for around one quarter of global CO_2 emissions [81], governments have been promoting the use of “green” vehicles in order to reduce pollution. Through various incentives, drivers are encouraged to replace conventional gasoline vehicles with electric vehicles (EVs) fully powered by electricity, hybrid cars using both electricity and internal combustion engine (ICE) or hydrogen fuel cell vehicles. Although the number of EV users has been continuously increasing in recent years, the pace of the global EV market growth remains slow. According to [187], vehicle purchase price, limited driving range and insufficient charger availability are the main barriers to wider EV adoption. The driving range of an EV, which is the maximum distance that

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Table 1
Characteristics of charging technologies available [18,137,162].

Charging technology	Typical charging time	Approximate installation cost
Level 1 Charger	4–11 h	800$
Level 2 Charger	1–4 h	1,000$–3,000$
Level 3 Charger	≤30 min	75,000$ for a 150 kW charger
Battery Swapping	5 min	500,000$ for BetterPlace swapping station in 2009

a vehicle can travel with a fully charged battery, is indeed a source of "range anxiety" for drivers. Many of them consider as a hassle the number of times they need to stop for recharging the battery during a trip and might be anxious about running out of charge before reaching their destination. Long waiting and recharging time may also dissuade drivers from adopting electric vehicles. However, with the development of new technologies, the EV offerings should improve in the future by producing more affordable cars with a longer driving range. The newest EV models already show longer driving range of around 200 miles with the highest performance provided by Tesla Model S (402 miles) [127] but the expansion of the EV market is still hindered by the lack of sufficient charging infrastructure in different parts of the world. The recent increase in EV sales has encouraged many governments especially in Europe and Asia to offer incentives for charging infrastructure projects. According to [143], plug-in vehicle sales in Europe have exploded in 2020, despite the overall decrease in the auto market due to the Covid19 pandemic crisis. Plug-in vehicles currently represent 13% of the auto market, including 6.5% for fully electric vehicles. In the United States, EV sales are expected to increase by around 70% in 2021 as compared to 2020 and to account for 3.55% of market share while it represented only 0.65% of the market in 2015 [129]. Accordingly the number of public EV chargers is expected to rise globally from 1 million in 2020 to 12 millions in 2040 [126]. However, as shown in Table 1, the installation of EV chargers, particularly fast ones, represent a large investment. Relocating charging stations is also a heavy and costly process as explained by Jochem et al. [86], hence, it is crucial that charging service providers select optimal locations for charging stations in a way to satisfy as much recharging demand as possible while meeting investment budget constraints. In this context, recharging demand is considered to be satisfied if drivers can complete their trip without running out of charge. The corresponding problem, called the charging station location problem thus represents the focus of our survey.

1.2. Overview of the literature

The charging station location problem (CSLP) belongs to the category of facility location problem (FLP) that has been widely studied in the literature (see [130] for a review on FLP). Due to its specificity, complexity and also its importance to promote faster EV adoption, an extensive literature has developed on the subject of charging station location. We thus believe that a comprehensive review of existing works might be helpful for researchers to understand the current challenges in modeling and solving the problem and identify new research perspectives. Since the number of related papers in the last two decades is large, we restricted our literature review to articles published in scientific journals only and involving mathematical programming approaches as well as explicit modeling of facility location variables. Moreover, even though most relevant articles have considered charging infrastructure planning for EVs, many works have investigated the case of refueling stations for other types of alternative fuel such as hydrogen. Therefore, we used different keywords in our survey to capture most relevant literature regardless of the type of energy used, examples include "electric vehicle charging station location'', "flow refueling location'', "alternative fuel vehicle station location'', etc. We searched in different academic databases including Google Scholar, Web Of Science and Scopus, which led to the identification of 179 research papers on CSLP. Figs. 1 and 2 respectively illustrate the evolution of CSLP publications over time and the journals in which they were published.

Figures show that the number of articles published has been significantly growing in the last seven years, with the highest number of papers being issued in years 2017 and 2018. Most papers belong to journals specialized in transportation such as Transportation Research Parts B, C, D and E as well as operations research journals including European Journal of Operational Research and Computers and Operations Research in addition to journals focused on energy management and sustainability issues such as Energy, International Journal of Hydrogen Energy and International Journal of Sustainable Transportation. This growing interest in the field has motivated the publication of several survey papers in the last five years. Table 2 illustrates the main focus elements of existing survey works based on literature discussions and classifications provided by the authors of these works.

To the best of our knowledge, the earliest survey in the field was proposed by Jing et al. [85] who examine the literature on network models for EVs, including CSLP, traffic assignment and vehicle routing models. Since CSLP is not the only focus of this paper, the number of CSLP articles considered is limited and no classification of the existing works is proposed. Ko et al. [102] discuss CSLP modeling features such as recharging demand representation and model constraints. They divide models into four categories according to their objective: set covering, maximal covering, p-center and p-median problems and propose a general classification table. They also review different approaches for estimating recharging demand. Pagany et al. [141] present different ways of literature classification such as model orientation (user-based, route-based and destination-based) and theoretical versus empirical approaches. A detailed review is proposed for empirical approaches only and focuses on data sources, model objectives, solution approaches and case-study regions. The surveys presented in [102] and [141] pertain to year 2016 or before while the most recent CSLP articles considered in [85] were published in 2015. Accordingly, these survey papers, yet focus on important aspects of existing works, do not capture the growing literature in the last four years (note here that 55% of the articles considered in our

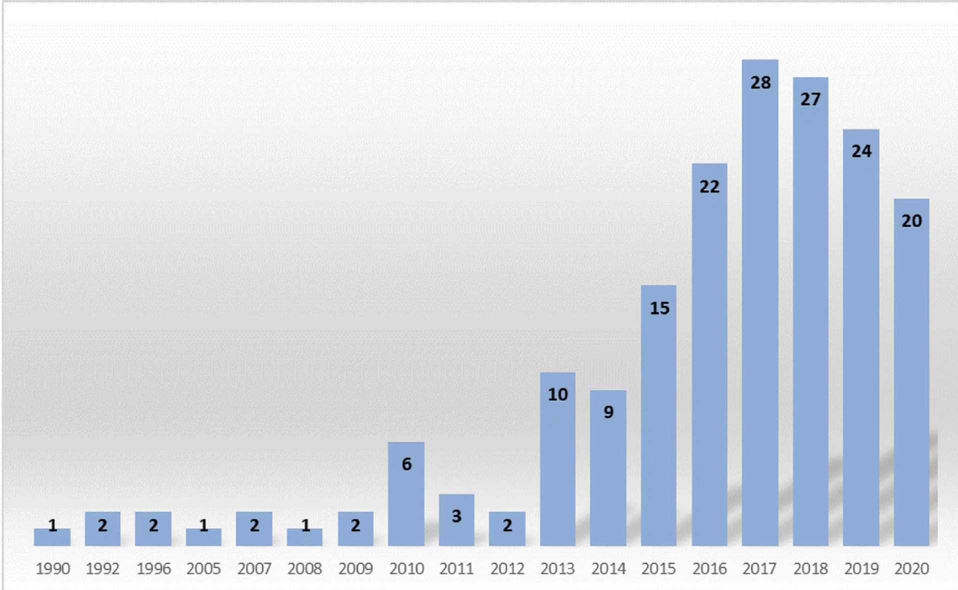

Fig. 1. CSLP publications over time.

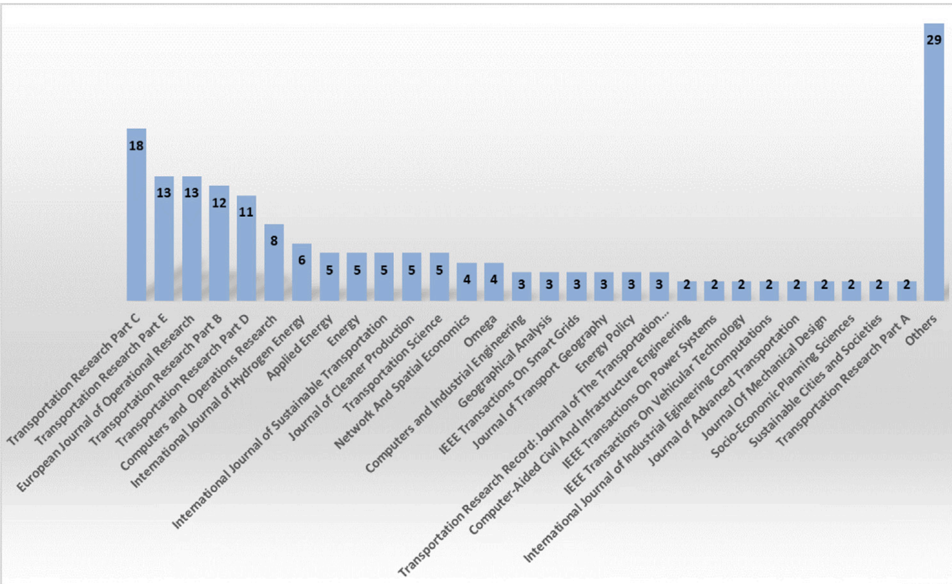

Fig. 2. Journals of publication of CSLP articles.

review were published between 2017 and 2020). More recently, other survey papers were proposed by Bilal and Rizwan [15], Lin et al. [122] and Shen et al. [159]. Bilal and Rizwan [15] first describe the charging infrastructure planning scenario in India then review possible objective functions, decision variables, constraints and solution techniques for CSLP models. They also examine the use of demand response programs (DRPs) to avoid high charging load on the power grid during peak hours through charging price variation and incentives. Shen et al. [159] consider flow-based and network equilibrium approaches and propose a classification of existing works according to objective function, model type and solution method. However, the scope of the literature reviewed is broad and involves three fields: planning for charging infrastructure, EV charging operations and public policy and business models, hence, only a restricted number of CSLP articles is presented by the authors. Finally, Lin et al. [122] conduct a brief survey on locating hydrogen refueling stations. They focus on typical mathematical formulations of the problem including set covering, maximal covering, p-median, flow-capturing and flow-refueling location models but they consider a small number of representative papers and do not classify them.

Table 2
Overview of survey articles on CSLP.

Survey element	Bilal and Rizwan [15]	Jing et al. [85]	Ko et al. [102]	Lin et al. [122]	Pagany et al. [141]	Shen et al. [159]	Our survey
Year	2020	2016	2017	2020	2019	2019	**2021**
#CSLP references[a]	<50	<50	<50	<50	119	<50	**179**
Review time range[b]	Until 2019	Until 2015	Until 2016	Until 2017	Until 2016	Until 2018	**Until 2020**
Flow/Node distinction			✓		✓	✓	✓
Objective functions	✓		✓	✓	✓	✓	✓
Decision variables	✓						✓
Coverage approaches			✓	✓		✓	✓
Typical formulations				✓			✓
Bi-level CSLP		✓					✓
Game theory						✓	✓
Solution methods	✓				✓	✓	✓
Case-studies			✓		✓		✓

[a]Number of references studying CSLP models.
[b]Time range considers the most recent CSLP models cited in the paper.

1.3. Contributions

In the present paper, we aim at providing a comprehensive review of CSLP studies published to date, in order to capture the various approaches proposed to model the problem, solve it efficiently and apply models in real-life. Based on our comprehensive survey, we then highlight discussion points related to CSLP and identify perspectives for future research in the field. The contributions of the present paper can be summarized as follows. First, we carry out a detailed analysis of the main elements and features of the CSLP. Namely, we investigate the way of modeling the recharging demand and how it can be fulfilled, particularly considering that driving range restriction for EVs represents one of the main differences between CSLP models and classical FLP approaches. Accordingly, we distinguish between conventional node-based models that assign customer demand to the nodes of a network and flow-based models that consider demand as a set of origin–destination (OD) trips taken by EV drivers. In the latter case, we classify the existing works according to their way of modeling the refueling of a trip and present typical mathematical formulations in each case. Furthermore, we discuss other elements of the CSLP including the model objective, structure (single-level or bi-level), decision variables, constraints, interaction between transportation and power distribution networks as well as dynamic and stochastic elements, which leads to different ways of classification of the literature. We also review and compare diverse methods adopted in the literature to efficiently solve the problem and handle computational difficulties arising from realistic features introduced in the model to enhance its practical relevancy. Finally, based on our detailed review, we discuss possible extensions of existing works and present an overall perspective on the long-term future research directions in the field.

Our survey is organized as follows. Section 2 investigates the main CSLP modeling issues. Then, solution methods and application of CSLP to real-life case-studies are discussed in Section 3. Classifications of existing works according to various criteria are proposed in Tables 3–12. Finally, Section 4 is devoted to discussing the findings of our survey and suggesting opportunities for future research while Section 5 presents some conclusions of our study.

2. Modeling the charging station location problem

In this section, we seek to review the main modeling issues addressed in the CSLP literature. First, we discuss different ways of representing the recharging demand and modeling its coverage then we outline some typical mathematical formulations of the problem. We also examine the use of game theoretical approaches to model interactions between decision makers involved in EV charging infrastructure planning. Moreover, we investigate different types of decisions integrated by researchers in CSLP models and discuss how they handle stochastic and dynamic components.

2.1. Recharging demand representation and coverage

2.1.1. Flow-based models

Contrary to classical facility location models that represent customer demand at the nodes of a network, a relatively new stream of the literature has been developed to address CSLP with flow-based demand. The idea here is that, due to a limited driving range, EV drivers who travel for long distances, typically exceeding their driving range, need to recharge their battery while on their way from their origin to their destination. The recharging demand is thus modeled as a set of origin–destination (OD) trips where fast CSs have to be adequately placed in a way that the distance between two consecutive stations does not exceed the driving range. This would allow drivers to travel from their origin to their destination and back without running out of charge. Since a growing body of literature has studied flow-based CSLP models, we propose here a detailed review of the main modeling aspects, namely characteristics such as bi-level structure, station capacity constraints, possibility of deviation from shortest paths as well as objective functions used and demand coverage approaches.

2.1.1.1. Model characteristics and objective function. In Table 3, we survey 77 papers dealing with flow-based models and classify them according to a number of modeling features. Due to the large number of surveyed papers, models that include uncertain and dynamic components will not be discussed in the current subsection but will rather be considered in detail in Section 2.4.

One of the first papers to introduce flow-based location models is Hodgson [62]. The author presents in this paper a flow capturing location model (FCLM) where the objective is to determine the location of a predetermined number of facilities p in a way to maximize the flow captured on the network. The FCLM is thus an extension of the maximal covering location model with the only difference lying in the definition of demand coverage. A flow on a given OD trip is considered covered or captured if there is at least one facility located along the trip. However, this model does not take into account the main constraint faced by EV drivers, that is the limited driving range. Kuby and Lim [105] hence introduce the flow refueling location model (FRLM) that considers EV driving range limitation. Namely, an EV flow is considered covered if an adequate number of CSs is placed on the corresponding trip in a way to allow an EV driver to travel from the origin of the trip to its destination and back without running out of charge. The FRLM as well as most papers surveyed in Table 3 consider vehicle flow coverage maximization as objective function. This might be explained by limited financial resources in EV charging infrastructure projects, hence, selecting optimal locations of a predetermined number of stations may be the best approach to achieve the highest flow coverage within a given budget. Other models use a set covering approach that focuses on cost minimization while ensuring full coverage of EV flows i.e., that all EV drivers can fulfill their journeys without running out of charge. Among the costs considered are those of CS building [47,54,77,178,181], energy storage systems [58], CS operation [58], battery investment [47,58,108], transformer investment [108], network expansion [183] and greenhouse emissions [212]. Cost measures might also include time components such as charging and waiting time [47,61,180] or travel time [25,125,183,212,216]. Cost minimization measures may be substituted by profit maximization measures [14,52] in case private investors are building the charging infrastructure and the profit generated from the recharging service is interrelated with station location decisions. Other objectives relate to service level [37,60,61,67,125,211], power or energy [186], environmental impact [212] and deviations from shortest paths [54,120,221].

Service level maximization can be represented using different approaches. For instance, Dong et al. [37] propose to minimize the total number of missed trips where a trip is considered missed if the battery state of charge becomes negative i.e., the available charge is not sufficient to reach the next CS. He et al. [61], Liu and Wang [125] and Zhang et al. [211] consider minimizing the cost of the inconvenience caused by missed trips. He et al. [60] model service through a customer utility function consisting of four components: travel time, availability of CSs, charging fees and attractiveness of CS location. Hong and Kuby [67] study a threshold coverage extension to the FRLM that focuses on the percentage of a zone's OD trips that can be successfully completed without running out of charge, given a typical driving range. The objective is to cluster covered flows to certain demand nodes, based on the idea that consumers will not buy a vehicle unless a certain threshold percentage of their trips can be completed. Furthermore, models combining a power distribution network and a transportation network may take into account power-related components in their objective functions such as minimizing power losses and voltage deviations of electricity distribution systems (see e.g. Wang et al. [186]). A few other models, such as those proposed by Guo et al. [54], Lin and Lin [120] and Zockaie et al. [221] seek to minimize EV driver deviations from their shortest paths. In these models, the authors do not consider the common assumption that EV drivers take the shortest path from their origin O to their destination D. Path selection is generally made among a set of predetermined paths from O to D, generated in a way that the detour from shortest ones does not exceed a certain tolerated level. In other cases, deviations are allowed but the amount of deviation is not included in the objective function since the focus is rather on flow coverage ([97] and [73]), travel time ([25], [69] and [216]), cost ([114]) or profit ([14]). It is also worth noticing here that most articles surveyed in Table 3 consider single objective models with the exception of four papers (Hodgson and Rosing [63], Wang et al. [186] and Wang and Wang [185]) that combine a flow coverage maximization objective with a second objective.

As shown in Table 3, it is common in the CSLP literature to assume unlimited CS capacity i.e., a large number of drivers can simultaneously recharge their battery at each CS. This assumption might be explained by the low market penetration of EVs in the early adoption phase of this technology as suggested in [77], which inevitably leads to a low utilization rate of the charging infrastructure. However, as explained in the introduction, the number of EV drivers is expected to significantly increase in the future, which may lead to higher pressure on governments to make the charging infrastructure available with the right capacity in order to allow drivers to complete their journeys without running out of charge and to avoid long queues waiting for recharging. One of the first papers to consider station capacity constraints in a flow-based CSLP is [172] who develop a capacitated extension of the FRLM introduced in [105], where the number of vehicles that can be refueled per time unit in each opened station should not exceed the station capacity. Wang [178], Wang and Lin [182] and Zheng and Peeta [217] also consider station capacity constraints based on the number of vehicles that can be served per time unit. Hosseini et al. [73] and Zockaie et al. [221] develop capacitated CSLP models where drivers are allowed to deviate from their shortest path on a given OD trip in a way to balance the charging load among the CSs opened on different paths. Hosseini and MirHassani [72] study an extension of the path-segment-based CSLP model introduced in [134], where capacity depends on the quantity of fuel needed in each station calculated as the distance between two stops multiplied by the rate of the fuel consumed. Other models that include CS capacity as decision variables and take into account the impact of these decisions on charging infrastructure cost (Ghamami et al. [47]), waiting time (Wang et al. [180]) and vehicle flow equilibrium (Wang et al. [184] and Wang et al. [180]) will be discussed in more detail in Section 2.3.

In addition to budget restrictions and station capacity constraints above discussed, some other types of constraints have been considered in the CSLP literature. Chung et al. [28], Kuby et al. [104] and Ngo et al. [136] include in their models equity constraints to ensure equitable access to charging stations among different demand regions. Chung et al. [28] use a flow-based form of equity constraint to ensure that the same percentage of flows is captured by refueling stations at each region. In [104], the authors point out that concentrating new liquefied natural gas stations in a few countries with high flow density may not be politically acceptable,

Table 3
Classification of flow-based models according to model characteristics and objective function.

Paper	Model characteristics			Minimization objective				Maximization objective		
	Bi-level	Station capacity	Deviation	Cost	Time	Deviation	Power	Cover.	Service	Profit
Arslan and Karasan [8]								✓		
Arslan et al. [9]			✓					✓		
Berman et al. [13]								✓		
Bernardo et al. [14]			✓							✓
Capar and Kuby [21]								✓		
Capar et al. [22]								✓		
Chen et al. [26]	✓	✓	✓	✓	✓					
Chen et al. [25]	✓		✓		✓					
Chung et al. [28]								✓		
Chung and Kwon [27]								✓		
Dong et al. [37]									✓	
Ghamami et al. [46]	✓	✓	✓	✓	✓					
Ghamami et al. [47]		✓		✓	✓					
Göpfert and Bock [51]			✓					✓		
Guo et al. [54]	✓		✓	✓		✓				
Guo et al. [52]										✓
He et al. [60]	✓			✓					✓	
He et al. [61]	✓		✓	✓	✓				✓	
He et al. [59]	✓		✓					✓		
He et al. [56]								✓		
He et al. [58]				✓						
Hodgson [62]								✓		
Hodgson and Rosing [63]								✓		
Hodgson et al. [64] [65]								✓		
Hong and Kuby [67]								✓	✓	
Honma and Kuby [69]			✓		✓					
Hosseini and MirHassani [72]		✓						✓		
Hosseini et al. [73]		✓	✓					✓		
Huang and Kockelman [76]	✓	✓	✓							✓
Huang et al. [77]			✓	✓						
Hwang et al. [78] [79]								✓		
Jeong [83]								✓		
Jochem et al. [86]								✓		
Kang et al. [91]								✓		
Kim and Kuby [97] [98]			✓					✓		
Kuby and Lim [105] [106]								✓		
Kuby et al. [104] [107]								✓		
Kunith et al. [108]				✓						
Kweon et al. [109]			✓					✓		
Li and Huang [114]			✓	✓						
Lim and Kuby [119]								✓		
Lin and Lin [120]	✓		✓			✓				
Liu and Wang [125]	✓(3l)		✓	✓	✓				✓	
MirHassani and Ebrazi [134]								✓		
Ngo et al. [136]	✓		✓		✓					
Nourbakhsh and Ouyang [138]				✓						
Riemann et al. [147]	✓		✓					✓		
Rose et al. [148]		✓		✓						
Shukla et al. [160]								✓		
Tran et al. [169]								✓		
Upchurch and Kuby [171]								✓		
Upchurch et al. [172]		✓						✓		
Ventura et al. [175]								✓		
Ventura et al. [176]				✓				✓		
Wang et al. [180]	✓	✓	✓	✓	✓					
Wang et al. [186]							✓	✓		
Wang et al. [183]	✓		✓	✓	✓					
Wang et al. [184]	✓	✓						✓		
Wang [178]		✓		✓						
Wang and Lin [181]				✓						
Wang and Lin [182]		✓		✓				✓		
Wang and Wang [185]				✓				✓		

(*continued on next page*)

therefore, they introduce and compare several side constraints to obtain a more equitable distribution of covered flows across European countries. They investigate transnational corridor constraints that require connectivity between different countries and

Table 3 (*continued*).

Paper	Model characteristics			Minimization objective				Maximization objective		
	Bi-level	Station capacity	Deviation	Cost	Time	Deviation	Power	Cover.	Service	Profit
Wen et al. [188]				✓				✓		
Xu and Meng [195]			✓					✓		
Xylia et al. [198]				✓						
Yildiz et al. [203]			✓					✓		
Yildiz et al. [204]				✓						
You and Hsieh [206]								✓		
Zhang et al. [211]		✓		✓					✓	
Zhang et al. [213]				✓						
Zhang et al. [212]	✓		✓	✓						
Zheng et al. [216]	✓		✓	✓	✓					
Zheng and Peeta [217]		✓		✓						
Zockaie et al. [221]		✓	✓			✓				

partial country coverage constraints that impose a minimum percentage coverage threshold in each country. Moreover, in relation to station capacity restrictions, service level constraints have been used in several works to guarantee customer satisfaction with regard to station availability and total time of waiting and recharging. For instance, Davidov and Pantoš [33] require the total time for fully recharging an EV battery to be lower than a certain limit, while Kunith et al. [108] limit the total charging time to the dwell time of a bus at each bus stop and Yang et al. [200] introduce a service quality constraint to ensure that the probability of an EV taxi being recharged at least once during the day is higher than a certain level. When the transportation network is coupled with the power distribution network, power flow related constraints should also be taken into consideration. Davidov and Pantoš's model [35] includes constraints that check the reliability of the electric power system, in order to maintain power flow balance after the placement of new CSs of different charging technologies. Similar power flow constraints ensuring secure operations of power systems are also considered in the path-segment-based model of Zhang et al. [211] and the FRLM with Load-Flow-Control studied by Scheiper et al. [152]. Finally, an example of side constraints related to station location restrictions is introduced by He et al. [56] to take into account nodes where charging stations already exist or where building stations is not permitted by local authorities.

Another feature of CSLP models considered in Table 3 is the bi-level modeling structure, an approach that has recently attracted a growing interest among researchers in the field. Such modeling structure is indeed required when the problem involves multiple decision makers with different objectives. In the CSLP case, we typically identify two main types of decision makers: on the one hand government or investors are those who build the charging infrastructure and on the other hand EV drivers are those who use the charging infrastructure. The reader is referred to Table 3 for examples of objectives considered by charging infrastructure planners in bi-level models and to Section 2.2 for the game theoretical context of bi-level models. It is however noticeable here that only five bi-level studies (Chen et al. [26], Ghamami et al. [46], Huang and Kockelman [76], Wang et al. [180] and Wang et al. [184]) take into account capacitated CSs to model the inconvenience resulting from insufficient capacity in users' objectives. Most of these models also allow deviations from shortest paths since drivers might prefer to deviate to avoid long queuing time in CSs.

2.1.1.2. Demand coverage approaches. Given the number of EV drivers traveling along each OD trip, the objective of flow-based models is to determine the best locations for CSs along these trips in a way to allow (a maximum of) EV drivers to complete their journey without running out of charge. Different approaches have thus been proposed in the literature to deal with the coverage of EV recharging demand i.e., ensuring that CSs are available to EV drivers whenever battery recharging is needed. We analyze here five of the most used methods in the literature to model recharging demand coverage, namely flow capturing, flow refueling, arc covering, path-segment and battery State-Of-Charge (SOC) tracking. Common assumptions made in the basic models include the following:

- A single path (typically the shortest) is used on each OD trip
- EV drivers carry round trips from their origin to their destination, therefore cyclically travel from O to D to O, etc., following the same path and accessing CSs in both directions
- Energy consumption rate is constant and equal for all vehicles, which allows to measure consumption as distance equivalent
- Energy consumption is symmetric i.e., the same power is needed in the outward and return directions of a trip
- All vehicles have the same battery capacity, or equivalently driving range. Note here that later extensions of the basic models e.g. by Arslan and Karasan [8], Hwang et al. [79], Guo et al. [53] and Liu and Wang [125] have incorporated vehicle classes with different values of the driving range.
- If there is a station at the origin of a trip, the vehicle will start with a full battery (tank), if not, the battery (tank) should be at least half full to allow the driver, in his return trip, to travel from the first station down the path to the origin and back without running out of charge
- Stations can be located at nodes of the network only and are uncapacitated

We illustrate in Fig. 3 a classification of the surveyed papers according to the coverage approach used and we explain in the following each approach then outline the basic mathematical model considered (parameter notations are summarized in Table 4).

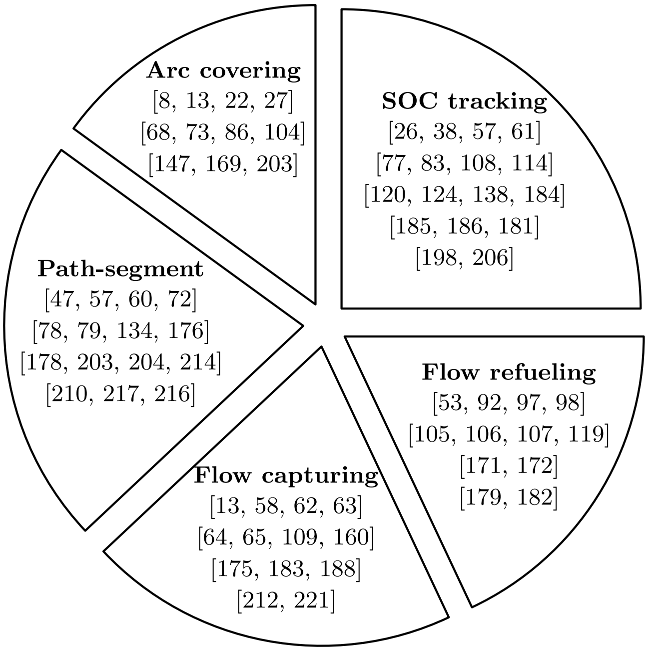

Fig. 3. Classification of flow-based models according to the demand coverage approach used.

Table 4
Model parameters.

Parameters	Description
$\mathcal{N}$	Set of nodes of the road network
$\mathcal{A}$	Set of arcs (i,j) of the road network, $i \in \mathcal{N}, j \in \mathcal{N}$
$\mathcal{Q}$	Set of origin–destination (OD) trips taken by EV drivers in the network
$\mathcal{M}$	Set of vehicles
$\mathcal{N}_q$	Set of nodes belonging to trip $q \in \mathcal{Q}$
$\hat{\mathcal{N}}_q$	Set of nodes of the expanded network belonging to trip q
$\mathcal{A}_q$	Set of directional arcs belonging to trip q
$\hat{\mathcal{A}}_q$	Set of directional arcs of the expanded network belonging to trip q
$\mathcal{Q}_i$	Subset of $\mathcal{Q}$ that contains all the paths passing node i
$\mathcal{H}$	Set of all combinations $(h \in \mathcal{H})$
h	Subscript introduced to represent a combination of facilities $k \in \mathcal{N}$ able to refuel one or more trips q
a_{hk}	A coefficient equal to 1 if node $k \in \mathcal{N}$ is in combination $h \in \mathcal{H}$ and 0 otherwise
b_{qh}	A coefficient equal to 1 if node combination h can refuel OD pair q and 0 otherwise
O_q, D_q	Respectively origin and destination of trip q
f_q	Flow of EVs on trip q
p	Number of CSs to be built
K_{ij}^q	Set of candidate sites, which can refuel the directional arc (i,j) on trip q
$d(i,j)$	Length of arc $(i,j) \in \mathcal{A}$
δ_{ijm}	Parameter equal to 1 if arc (i,j) is on a path used by vehicle $m \in \mathcal{M}$
c_i	Cost of building a charging station at node i
R	Range of electric vehicles (or equivalently maximum recharging amount)
L	Large value
Decision Variables	
x_k	Binary station location variable; $x_k = 1$ if a station is located at node k, $x_k = 0$ otherwise
y_q	Binary flow coverage variable; $y_q = 1$ if EV flow on trip q is covered, $y_q = 0$ otherwise
v_h	Binary refueling combination variable; $v_h = 1$ if stations are opened at all nodes belonging to combination h, $v_h = 0$ otherwise
y_{ij}^q	Continuous flow variable on an arc (i,j), which belongs to expanded network arcs of path q (i.e., $\hat{\mathcal{A}}_q$)
B_{im}	Continuous variable representing the remaining amount of charge (fuel) at node i for a vehicle m
R_{im}	Continuous variable representing the recharging amount at node i for a vehicle m
A_{im}	Continuous variable representing an adjustment coefficient for a vehicle m recharging at a node i
y_{im}	Binary recharging variable; $y_{im} = 1$ if a vehicle m is recharged at node i, $y_{im} = 0$ otherwise

2.1.1.2.1. Flow capturing The flow capturing location model (FCLM) was introduced by Hodgson [62], based on the well-known maximal covering facility location model introduced by Church and Revelle [29]. In the FCLM, a trip q is considered covered or captured if at least one CS is opened along q. The model includes binary variables x_k representing station location at every node

k and binary variables y_q representing trip q coverage decisions. The FCLM is formulated as follows:

$$Max\ Z_1 = \sum_{q \in Q} f_q y_q \tag{1}$$

$$s.t. \sum_{k \in \mathcal{N}_q} x_k \geq y_q \qquad \forall q \in Q \tag{2}$$

$$\sum_{k \in \mathcal{N}} x_k = p \tag{3}$$

$$x_k, y_q \in \{0,1\} \qquad \forall q \in Q, k \in \mathcal{N} \tag{4}$$

The objective function (1) maximizes the total EV flow that can be captured by opened stations. Constraints (2) ensure that a trip q is covered if at least one station is opened along q, while constraint (3) sets the number of open stations to p and constraints (4) are variable definition constraints. The main weakness of the FCLM is the fact that it does not take into account the limited driving range of EVs.

2.1.1.2.2. Flow refueling The flow refueling location model (FRLM) introduced by Kuby and Lim [105] has been the first flow-based location model to take into account driving range limitation for EVs (or more generally for alternative fuel vehicles). Given a fixed driving range R, trip coverage is defined using combinations of stations generated in a pre-processing step. A combination h can cover a trip q when an EV driver is able to travel from trip origin O_q to trip destination D_q and back without running out of charge by stopping to recharge at the stations belonging to combination h. A trip q is thus considered refueled or covered if at least one of the combinations capable of refueling q is used i.e., if CSs are opened at all nodes belonging to this combination. In addition to location variables x_k and trip coverage variables y_q, the model uses binary variables v_h indicating if stations are opened at all nodes of combination h. The FRLM is formulated as follows:

$$Max\ Z_1 = \sum_{q \in Q} f_q y_q \tag{5}$$

$$s.t. \sum_{h \in \mathcal{H}} b_{qh} v_h \geq y_q \qquad \forall q \in Q \tag{6}$$

$$a_{hk} x_k \geq v_h \qquad \forall h \in \mathcal{H}, k \in \mathcal{N} | a_{hk} = 1 \tag{7}$$

$$\sum_{k \in \mathcal{N}} x_k = p \tag{8}$$

$$x_k, y_q, v_h \in \{0,1\} \qquad \forall q \in Q, k \in \mathcal{N}, h \in \mathcal{H} \tag{9}$$

The objective function (5) maximizes the total EV flow covered. Constraints (6) state that a trip q is refueled if at least one combination of facilities h able to refuel q is opened ($v_h = 1$ and $b_{qh} = 1$ in this case). Constraints (7) set v_h to zero unless all the facilities in combination h are open. Constraint (8) requires exactly p facilities to be built and constraints (9) are variable definition constraints. The main shortcoming of the FRLM is that the generation of combinations is time-consuming for real-life instances and it was proven by Capar and Kuby [21] to grow exponentially as the number of nodes increases along a path.

2.1.1.2.3. Arc covering To overcome the difficulty related to generating combinations in the FRLM, Capar et al. [22] propose a new formulation of the problem based on an arc covering approach, according to which a trip q is considered covered if each arc belonging to q is covered. The key innovation of this new formulation is the introduction of K_{ij}^q, set of nodes that can cover arc (i, j) on trip q. K_{ij}^q can be defined as the set of candidate nodes where completely recharging the battery allows the driver to cross the portion of the route from i to j without running out of charge. The problem is formulated as follows:

$$Max\ Z_1 = \sum_{q \in Q} f_q y_q \tag{10}$$

$$s.t. \sum_{k \in K_{ij}^q} x_k \geq y_q \qquad \forall q \in Q, (i,j) \in \mathcal{A}_q \tag{11}$$

$$\sum_{k \in \mathcal{N}} x_k = p \tag{12}$$

$$x_k, y_q \in \{0,1\} \qquad \forall q \in Q, k \in \mathcal{N} \tag{13}$$

The objective function (10) maximizes the total EV flow covered. Constraints (11) ensure trip q coverage if for every arc (i, j) belonging to q there is at least an open station at one of the nodes of set K_{ij}^q, which is the set of candidate nodes k that allow an EV driver to drive across the entire distance from node i to node j if he completely recharges the vehicle with range R at k. Similarly to previous models, constraint (12) and (13) respectively set the number of stations to be built and define decision variables.

2.1.1.2.4. Path-segment A path-segment $[i, j]$ of a trip q is defined as a portion of the path used to travel from the origin of the trip O_q to its destination D_q. It is composed of a set of consecutive arcs to be crossed on q in order to travel from node i to node j. The length of a path-segment is thus the sum of the lengths of the arcs composing it. The idea of a path-segment-based coverage approach is to decompose each trip q into a sequence of path-segments that are consecutively traversed when traveling from O_q to D_q. In order for trip q to be covered, the length of each of these path-segments should not exceed the driving range R (or $R/2$ if the origin of the segment corresponds to O_q or its destination corresponds to D_q) and a charging station should be placed at the origin

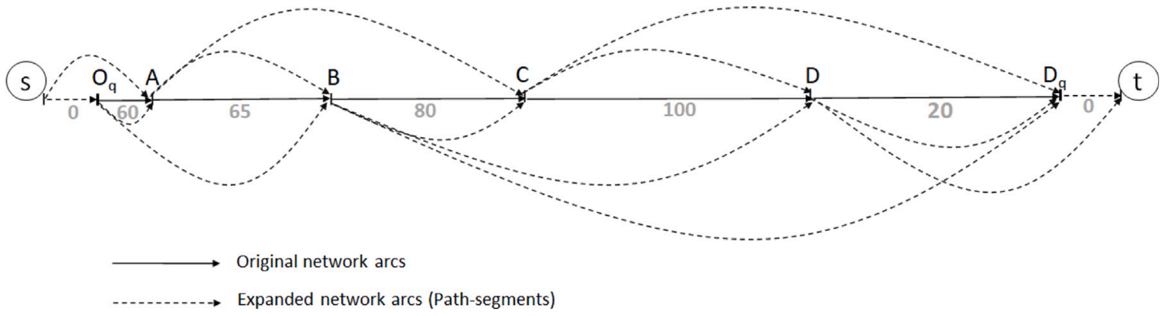

Fig. 4. Example of MirHassani and Ebrazi's [134] expanded network on a trip from origin O_q to destination D_q with $R = 200$.

node of each path-segment built (except if this origin node corresponds to O_q). This allows EV drivers to recharge their batteries at each of these stations and reach their destination without running out of charge.

The notion of path-segment was defined and used by Yildiz et al. [203] in their deviation-based CSLP model. Equivalently, de Vries and Duijzer [36] introduced the concept of cycle segment, which was also adopted in Kchaou-Boujelben and Gicquel [94] to develop a more efficient MILP formulation for the problem. However, one of the first works to introduce the idea of path-segment-based coverage is MirHassani and Ebrazi [134]. Although their method does not explicitly use the notion of path-segment, their model based on the concept of expanded network implicitly involves building path-segments. More precisely, an expanded network $(\hat{\mathcal{N}}_q, \hat{\mathcal{A}}_q)$ is constructed for each trip q in the following way: the set of nodes $\hat{\mathcal{N}}_q$ contains the original set of nodes (N_q) in addition to artificial source s and sink t, while the set of arcs $\hat{\mathcal{A}}_q$ is obtained by connecting any two nodes of $\hat{\mathcal{N}}_q$ if the length of the path-segment linking them does not exceed the range R (or $R/2$ if we consider round trip to the origin or to the destination nodes).

The corresponding mathematical model presented below aims at maximizing the EV flow covered by determining which trips should be covered and building an optimal path on each covered trip while placing a station at each node through which a path is built. The objective function (14) maximizes the flow covered, with variable y^q_{st} indicating if the flow on trip q is covered or not. Constraints (15) are mass balance constraints while constraints (16) ensure that a station is opened at any node through which a path is constructed. Constraint (17) limits the number of stations to p and constraints (18)–(19) are variable definition constraints.

$$Max\ Z_1 = \sum_{q \in Q} f_q (1 - y^q_{st}) \tag{14}$$

$$s.t. \sum_{j|(i,j)\in\hat{\mathcal{A}}_q} y^q_{ij} - \sum_{j|(j,i)\in\hat{\mathcal{A}}_q} y^q_{ji} = \begin{cases} 1 & i = s \\ -1 & i = t \\ 0 & i \neq s,t \end{cases} \qquad \forall q \in Q, i \in \hat{\mathcal{N}}_q \tag{15}$$

$$\sum_{j|(j,i)\in\hat{\mathcal{A}}_q} y^q_{ji} \leq x_i \qquad \forall i \in \mathcal{N}, q \in Q_i \tag{16}$$

$$\sum_{k\in\mathcal{N}} x_k = p \tag{17}$$

$$y^q_{ij} \geq 0 \qquad \forall q \in Q, (i,j) \in \hat{\mathcal{A}}_q \tag{18}$$

$$x_k \in \{0,1\} \qquad \forall k \in \mathcal{N} \tag{19}$$

To illustrate MirHassani and Ebrazi's [134] expanded network concept, an example is given in Fig. 4 with a driving range set to 200. In this case, the expanded network involves arcs $[s, O_q], [s, A], [O_q, A], [O_q, B], [A, B], [A, C], [B, C], [B, D], [B, D_q], [C, D], [C, D_q], [D, D_q], [D, t]$ and $[D_q, t]$. Note that only arcs $[s, O_q]$ and $[s, A]$ (respectively arcs $[D_q, t]$ and $[D, t]$) are connected to the source s (respectively to the sink t) because in this case the length of the arc has to be shorter than 100 (i.e, $R/2$) to ensure a round trip to the origin (respectively to the destination) without running out of charge. In all the other cases, the maximum length of the arcs of the expanded network should be 200 (i.e, R).

2.1.1.2.5. Battery SOC tracking Wang and Lin [181] propose a set covering formulation where the battery state-of-charge (SOC) is tracked at each node visited by the vehicle, in a way to ensure that the remaining amount of charge stays positive. Variable B_{im} represents the amount of charge remaining at node i for vehicle m and is evaluated as follows:

(1) if i is the origin of a trip, B_{im} is set to the maximum recharging amount R

(2) if i is not an origin and vehicle m traveled from node j to node i, B_{im} is equal to the remaining amount of charge at node j B_{jm} plus the amount recharged at node j R_{jm} minus the amount of charge consumed to travel from j to i, equivalently distance $d(j,i)$.

This is very similar to the logic used in lot-sizing models to describe inventory level change from a time period to the next. The model also involves continuous variable A_{im} representing an adjustment coefficient for a vehicle m being recharged at node i and binary variable y_{im} indicating if a vehicle m is recharged at node i. The problem is formulated as follows:

$$Min\ Z_2 = \sum_{i\in\mathcal{N}} c_i x_i \tag{20}$$

$$s.t.\ B_{im} \geq 0 \qquad \forall i \in \mathcal{N}, m \in \mathcal{M} \tag{21}$$

$$B_{im} = B_{jm} + R_{jm} - d(j,i) \times \delta_{jim} \qquad \forall (j,i) \in \mathcal{A}, m \in \mathcal{M} \tag{22}$$

$$R_{im} \leq R - B_{im} \qquad \forall i \in \mathcal{N}, m \in \mathcal{M} \tag{23}$$

$$R_{im} = y_{im} R - A_{im} \qquad \forall i \in \mathcal{N}, m \in \mathcal{M} \tag{24}$$

$$\sum_{m\in\mathcal{M}} y_{im} \leq L x_i \qquad \forall i \in \mathcal{N} \tag{25}$$

$$x_i \in \{0,1\} \qquad \forall i \in \mathcal{N} \tag{26}$$

$$y_{im} \in \{0,1\} \qquad \forall i \in \mathcal{N}, m \in \mathcal{M} \tag{27}$$

$$A_{im}, B_{im}, R_{im} \geq 0 \qquad \forall i \in \mathcal{N}, m \in \mathcal{M} \tag{28}$$

The objective (20) is to minimize the fixed costs of building charging stations. Constraints (21) make sure that when a vehicle m arrives at a site i, its remaining charge is greater than or equal to zero, in other words, the amount of charge in the previous site j must be greater than or equal to the charge consumed during the distance traveled from j to i. Constraints (22) define for each visited arc (j,i) the relationship between B_{im} (the remaining amount of charge at node i), B_{jm} (the remaining amount of charge at node j), R_{jm} (the amount recharged at node j) and the charge consumed to travel from j to i, with $d(j,i)$ being the distance from j to i and δ_{jim} being a parameter equal to 1 if arc (j,i) is on a trip used by vehicle m. Constraints (23) state that the recharging amount at a node i is less than or equal to the maximum recharging amount (driving range R) minus the remaining charge at node i. Constraints (24) set the recharging amount at a node i to the maximum recharging amount minus an adjustment coefficient A_{im} used to calibrate the amount recharged at each node. Constraints (25) and (26)–(28) respectively model the relationship between location and recharging variables and define decision variables.

2.1.2. Other models

Another part of the literature on CSLP has considered classical facility location modeling approaches (see e.g. Klose and Drexl [100]) where the recharging demand does not depend on EV trips but simply arises at the nodes of a road network. The node-based demand usually originates from a cluster of EV drivers who prefer to recharge their battery in a station located close to their home, workplace or any service facility such as a shopping mall. This situation thus usually corresponds to the use of slow chargers where the vehicle recharging might take many hours while the vehicle user is resting at home, working or shopping. A classification of the related papers according to the model type, CS capacity constraint and objective function is shown in Table 5. We can see from the table that a significant number of papers in this category use a location–allocation (LA) approach that minimizes total system costs while satisfying all customer demand. The system costs generally include traveling costs to charging stations and fixed investment costs as well as costs related to the power grid network that have been considered by a number of papers such as Cui et al. [31], Islam et al. [82] and Sadeghi-Barzani et al. [149].

Other works propose to use a set covering (Set-Cover) approach ensuring that every customer can access a charging facility situated within a given service distance from their location, while minimizing investment costs. The introduction of the service distance threshold is actually a way to model the EV driving range or alternatively a reasonable walking distance or driving time. For instance, Stephens-Romero et al. [163] analyze service coverage provided by hydrogen fueling stations for a maximum driving time set to 5 min, while Lam et al. [110] impose in their model that for every demand node, the total charging capacity accessible within a given distance is higher than a certain minimum threshold. Huang et al. [75] consider two groups of drivers, where the service distance for the first group is set to a maximum walking distance from an activity location such as a shopping or dining center and the threshold is set for the second group to a driving distance before the battery is depleted, as this group is assumed to conduct longer trips. However, when it is not possible to cover all customer demand due to budget constraints, a maximal covering (Max-Cover) approach can be used. He et al. [57] compare set covering, maximal covering and p-median station location approaches and conclude that p-median would allow more convenient access to charging stations for communities with higher demand.

Furthermore, multi-objective models have been proposed in the node-based CSLP literature (see e.g. Bai et al. [11], Brey et al. [19], Chen et al. [24], Hodgson and Rosing [63] and Yi and Bauer [202]). For instance, in order to assign higher weight to candidate locations having more vehicle kilometers traveled within their range, Brey et al. [19] consider maximizing traffic intensity captured by charging stations together with a p-median objective minimizing the average distance traveled to refuel vehicles. Hodgson and Rosing [63] and Yi and Bauer [202] combine demand coverage maximization and facility access cost minimization while Bai et al. [11] and Chen et al. [24] determine the optimal locations of battery swapping stations and battery charging stations in a way to minimize system costs and maximize service level, either by maximizing the number of customers served [24] or by minimizing the time spent by drivers to complete recharging [11].

Flow-based and node-based demand representations are not the only approaches used in CSLP models. A different method based on vehicle trajectory has been adopted by Jung et al. [88], Liu et al. [124], Shahraki et al. [158], Tu et al. [170] and Yang et al. [200] to solve the CSLP for electric taxis. Jung et al. [88] develop a stochastic itinerary interception model under passenger demand uncertainty. The trajectory of each vehicle is dynamically determined by the model according to the occurring passenger demand

Table 5
Classification of node-based models according to model characteristics and objective function.

Paper	Model characteristics		Minimize cost			Maximization objective
	Model approach	Station capacity	Investment cost	Travel to CS	Power network related	Demand coverage
Asamer et al. [10]	Max-Cover					✓
Baouche et al. [12]	LA	✓	✓	✓		
Bouguerra and Bhar Layeb [16]	Set-Cover	✓	✓	✓		
Cavadas et al. [23]	Max-Cover	✓				✓
Cui et al. [31]	LA	✓	✓		✓	
Dong et al. [38]	Max-Cover					✓
Frade et al. [43]	Max-Cover	✓				✓
Gimenez-Gaydou et al. [49]	Max-Cover					✓
Huang et al. [75]	Max-Cover					✓
Hu et al. [74]	Max-Cover	✓				✓
Islam et al. [82]	LA	✓	✓	✓	✓	
Kang et al. [92]	LA			✓		
Ko and Shim [103]	LA	✓		✓		
Lam et al. [110]	Set-Cover		✓			
Lin et al. [121]	LA			✓		
Ogden and Nicholas [139]	LA			✓		
Sadeghi-Barzani et al. [149]	LA	✓	✓	✓	✓	
Stephens-Romero et al. [163]	Set-Cover		✓			
Vazifeh et al. [174]	Set-Cover		✓	✓		
Xie et al. [193]	Set-Cover	✓	✓			
Xu and He [194]	LA		✓	✓		
Xu et al. [196]	LA			✓		
Zhu et al. [220]	LA	✓	✓	✓		

and the recharging needs of the vehicle, which also allows the model to optimize the siting and sizing of charging stations. Liu et al. [124] and Shahraki et al. [158] consider problems where the chain of OD trips traveled by each Plug-in Hybrid EV is known but the decision of recharging the battery at the end of each trip has to be made by the model or alternatively gasoline energy should be used if recharging is not possible. Tu et al. [170] use actual information on conventional taxi trajectories and passenger demand extracted from real-life large volume GPS data to build a spatial–temporal model that optimizes the location of charging stations in a way to maximize the total distance traveled by electric taxis and minimize the total waiting time at charging stations under station capacity constraints. Yang et al. [200] consider the dwell pattern of taxis i.e., the succession of nodes where taxi drivers would stop for rest and use queueing theory to estimate the probability of taxis being charged at their dwell locations. Stations are located in a way to minimize costs and achieve a certain probability that a taxi can be recharged at least once a day in one of the dwell locations.

Location-routing problems have also been investigated for electric vehicles in order to determine optimal routes and CS locations that enable the vehicles to satisfy customer delivery requests without running out of charge. Given customer demand at different nodes of the network and considering a limited driving range for vehicles, most models in this category involve battery SOC tracking constraints (see Section 2.1.1.2.5) to evaluate the remaining energy at each portion of a candidate route and thus decide at each node of the network whether to locate a charging station [6,66,117,118,142,154,156,157,201,208,219]. Finally, it is also possible to investigate CSLP in situations where information on recharging demand is not available. For instance, Arkin et al. [7] and Gagarin and Corcoran [44] use graph theory to place a minimum number of battery charging stations on a road network. The objective in Arkin et al. [7] is to enable drivers to move between any two nodes of the network using routes that are as much close as possible to the shortest paths while in Gagarin and Corcoran [44], we seek to allow drivers find a number of possible recharging options within their driving range distance .

2.2. Game theory approaches

As explained in Gibbons [48] and Osborne [140], the objective of game theory is to model situations where multiple decision makers interact according to certain rules then receive payoffs that depend on the decisions they made. The first group of decision makers in CSLPs involve CS location planners, namely governments and private institutions while the second group of decision makers consists of users of the charging infrastructure i.e., the drivers of vehicles requiring recharging service. Private investors aim at satisfying customers while maximizing their profit whereas the objective of governmental institutions is to minimize total social cost and promote an equitable access to recharging facilities. EV drivers need to accomplish their trips with the least time and cost but may also consider other factors such as the attractiveness of the charging locations. Competitive interaction among entrants in the EV CS market has been considered by Bernardo et al. [14] and Guo et al. [52]. Both works study equilibrium location and pricing in situations where CS investors aim to maximize their profit and EV drivers need to maximize their utility. The utility function considered in Bernardo et al. [14] takes into account deviation from the shortest path when traveling from the origin to the destination of a trip as well as pricing and existence of amenities at the charging location. In addition to destination attractiveness and charging cost, the utility function used in Guo et al. [52] also includes travel time and total capacity available at the charging location.

The review of the CSLP literature related to game theory shows a growing interest in the field, in particular in the development of bi-level flow-based models, where a CSLP is considered in the upper level and a network flow equilibrium problem is solved in the lower level. The difference between the objective of network planners (upper level) and the one of network users (lower level) motivates the splitting of the problem into two stages. Instead of assuming that users will follow the optimal setting decided by planners in terms of selected path and recharging plan, bi-level approaches incorporate the selfish behavior of drivers in the decision process. EV drivers would usually select the path that allows them to travel from their origin to their destination without running out of charge in a way to minimize their own travel time or distance [25,59,61,85,112,117,125,133,147,180,183,212,216]. Travel time obviously includes driving time but also recharging time [59,61,84,125] and waiting time in CSs [180]. Moreover, in practice, travel time on a portion of the road depends on the flow of vehicles crossing that portion. This aspect has been considered in the literature through modeling the relationship between the travel time and the flow. A linear relationship has been used by Chen et al. [25] while a non-linear relationship based on the Bureau of Public Roads (BPR) function BPR [17] has been adopted by Miralinaghi et al. [133], Riemann et al. [147], Wang et al. [183], Zhang et al. [212] and Zheng et al. [216]. In addition to travel time, drivers might consider other factors to select their paths and recharging plans. For instance, He et al. [60] suggest that drivers make their decision based on a utility function that includes travel time, availability of CSs, charging cost and location attractiveness, while Wang et al. [184] use a utility function equal to the probability of drivers to be served in case charging facilities have limited capacity. In terms of modeling structure, the upper and lower levels of bi-level CSLPs are generally linked through complementary constraints derived from network flow equilibrium conditions in the lower level problem and added to the upper level problem (see e.g. Chen et al. [25] and He et al. [60]).

In addition, it is worth noticing here that the bi-level CSLP models cited above assume in the lower level that drivers select their paths among a set of predetermined paths while in other situations (see e.g. Li et al. [117] and Lin and Lin [120]), the lower level might involve solving EV routing problems where each driver determines, based on the CSs located by network planners, the shortest path that allows him to complete his trip without running out of charge. Another noticeable work proposing a different modeling structure is the one presented by Liu and Wang [125]. The authors develop a tri-level approach where in addition to location decisions made in the upper level and path selection decisions made in the lower level, users select the type of technology for their vehicles, among traditional plug-in technology or wireless recharging technology.

2.3. Decision variables

CSLP models are mainly intended to determine the best locations for CSs. However, other decisions related to the recharging plan of EVs and to trip coverage are necessary to evaluate the extent of recharging demand satisfaction in flow-based CSLP models, while node-based models generally use customer-station assignment variables to ensure demand satisfaction. Additional decisions such as CS capacity (number of chargers in each opened station), technology choice (battery swapping or recharging, fast or slow recharging), deviation from shortest path and vehicle routing decisions are also important to incorporate in the problem modeling in order to capture interrelations with the main decisions of the problem. Table 6 shows a survey of the models that include one or more of these decisions together with the model type (flow-based, node-based or other). According to the table, deviation from shortest path and station capacity are among the most popular decisions considered in CSLP models while less attention has been paid to technology choice and integrated location-routing decisions.

2.3.1. Capacity decisions

The capacitated node-based facility location problem has been widely studied in the literature (see e.g. Wu et al. [190]) but we are here more interested in its adaptation to the EV context. For instance, Cui et al. [31], Rajabi-Ghahnavieh and Sadeghi-Barzani [145] and Sadeghi-Barzani et al. [149] investigate the interrelation between station location, station capacity and electric network related decisions through mixed integer non linear programs (MINLP) that integrate a transportation network and an electric network. Bai et al. [11] combine station location and capacity decisions with technology selection (swapping or charging stations) in a bi-objective model studying the trade-off between cost minimization and quality of service maximization, the latter being evaluated as the time spent by a driver to complete charging. Xie et al. [193] develop a two-stage approach where CSs are sited in the first level and capacities of photovoltaic panels and storage units in each opened station are determined in the second level. In general, increasing charging station capacity by building more chargers in each station means more coverage of the demand but leads to higher investment cost. For example, Zhang et al. [210] Zhang et al. [211] and Xie et al. [192] study such trade-off by incorporating the cost of station installation and the penalty for unsatisfied demand in their objective functions. Moreover, another advantage of higher station capacity is a better service offered to EV drivers and a reduction of their waiting time in front of CSs. The cost of waiting time is evaluated by Dong et al. [39] and Ghamami et al. [47] using queuing theory and included in their objective functions. Xie et al. [192] and Yang et al. [200] use a probabilistic approach to derive service quality constraints: Xie et al. [192] incorporate in their model a stochastic chance constraint imposing that the probability of an EV driver to find a vacant charger within a short waiting time should be high enough while Yang et al. [200] consider that the probability that an electric taxi can be recharged at least once during the day is higher than a certain level.

However, one should be aware that increasing capacity in CSs causes more pressure on the electric network. This requires to examine the trade-off between the gain in demand coverage or waiting time and the negative impact on the power distribution network, as pointed out by Wang et al. [186]. That said, a reduced investment budget might result in insufficient capacity in some stations and thus forces drivers to use different recharging plans. In this case, drivers must determine, based on the location and size of opened stations, which combination of stations to be used to travel on each OD trip in order to complete their journey without

Table 6
Classification of papers according to problem decision variables.

	Model type	Station capacity	Technology choice	Deviation	Routing
An [2]	Node	✓			
Anjos et al. [3]	Node/Flow	✓		✓	
Arias et al. [6]	Other				✓
Bai et al. [11]	Node	✓	✓		
Bouguerra and Bhar Layeb [16]	Node	✓			
Chen et al. [26]	Flow	✓		✓	
Chen et al. [24]	Node		✓		
Chen et al. [25]	Flow			✓	
Cui et al. [31]	Node	✓			
Davidov and Pantoš [34] [35]	Flow		✓		
Dong et al. [37]	Flow		✓		
Dong et al. [39]	Flow	✓			
Ghamami et al. [46]	Flow			✓	
Ghamami et al. [47]	Flow	✓	✓		
Guo et al. [54]	Flow			✓	
Han et al. [55]	Other	✓			
He et al. [61], [59]	Flow			✓	
He et al. [58]	Flow		✓		
Hof et al. [66]	Other				✓
Hong et al. [68]	Other				✓
Hosseini and MirHassani [71]	Flow	✓			
Hosseini et al. [73]	Flow			✓	
Huang and Kockelman [76]	Flow	✓		✓	
Huang et al. [77]	Flow			✓	
Islam et al. [82]	Node	✓			
Jung et al. [88]	Other	✓			
Kabli et al. [89]	Node	✓			
Kang and Recker [93]	Other				✓
Kim and Kuby [97] [98]	Flow			✓	
Kınay et al. [99]	Flow			✓	
Kweon et al. [109]	Flow			✓	
Lee et al. [112]	Flow			✓	
Li and Huang [114]	Flow			✓	
Li et al. [115]	Flow			✓	
Li et al. [117]	Flow				✓
Lin and Lin [120]	Flow			✓	
Lin et al. [123]	Flow	✓			
Li-Ying and Yuan-Bin [118]	Other		✓		✓
Liu and Wang [125]	Flow		✓	✓	
Miralinaghi et al. [133]	Flow			✓	
Ngo et al. [136]	Flow			✓	
Paz et al. [142]	Other		✓		✓
Quddus et al. [144]	Node	✓	✓		
Rajabi-Ghahnavieh and Sadeghi-Barzani [145]	Node	✓			
Ribeiro et al. [146]	Other				✓
Riemann et al. [147]	Flow			✓	
Sadeghi-Barzani et al. [149]	Node	✓			
Schiffer et al. [154]	Other				✓
Schiffer and Walther [155] [156] [157]	Other				✓
Shahraki et al. [158]	Other		✓		
Tafakkori et al. [166]	Flow	✓	✓		
Tao et al. [167]	Flow			✓	
Wang et al. [180]	Flow	✓		✓	
Wang et al. [186] [184]	Flow	✓			
Wang et al. [179]	Flow		✓	✓	
Wang et al. [183]	Flow			✓	
Wang and Lin [182]	Flow		✓		
Xi et al. [191]	Flow		✓		
Xie et al. [192]	Flow	✓		✓	
Xie et al. [193]	Node	✓			
Xu and Meng [195]	Flow			✓	

(*continued on next page*)

running out of charge (see e.g. Wang et al. [184]). Drivers might also be forced to take a path longer than their shortest path in order for them to find vacant chargers. This situation is represented by Wang et al. [180] and Xie et al. [192] using integrated CSLP models combining station capacity decisions and path selection decisions. Other integrated models incorporate technology selection

Table 6 (*continued*).

	Model type	Station capacity	Technology choice	Deviation	Routing
Xu et al. [197]	Flow			✓	
Xylia et al. [198]	Flow		✓		
Yang et al. [200]	Other	✓			
Yang and Sun [201]	Other				✓
Yang [199]	Flow	✓		✓	
Yildiz et al. [203] [205]	Flow			✓	
You and Hsieh [206]	Flow		✓		
Zhang et al. [210] [211]	Flow	✓			
Zhang et al. [208]	Other				✓
Zhang et al. [212]	Flow		✓	✓	
Zheng et al. [216]	Flow			✓	
Zhou et al. [218]	Flow	✓		✓	
Zhou and Tan [219]	Other				✓
Zhu et al. [220]	Node	✓			
Zockaie et al. [221]	Flow			✓	

with siting and sizing of CSs in a way to analyze possible trade-offs. For instance, Ghamami et al. [47] consider the EV battery technology as a decision variable in their model since a higher investment in premium battery technology would lead to a longer driving range and so to less recharging demand and shorter waiting time in CSs.

With regard to capacity constraint modeling, a relevant question here is how short-term charging capacity management that depends in practice on an "operational" measure of occupancy can be combined with long-term station location decisions in a same model. If we consider general fixed-charge capacitated location problems (see e.g. Klose and Gortz [101] and Wu et al. [190]), we can notice that capacity constraints simply compare the total volume assigned to a facility with the maximum volume that the facility can handle during the planning period. Obviously, since facility location is a strategic problem, the planning period is generally of one or many years, thus, the capacity constraints used in the model are only aggregate representation of the needs in each location that help in facility sizing but cannot be used in day-to-day capacity management. However, once facilities are built, it is possible to adjust customer assignment decisions in order to transfer demand from a congested location to another location with idle capacity.

In the CSLP literature, similar concerns exist on how to model capacity constraints in practice. Most authors do not clearly explain how station capacity can be measured in a strategic planning horizon. Ghamami et al. [47] consider station congestion through the introduction of average waiting time cost in the objective function estimated based on the total capacity of the station and the total number of EVs charging at the station during the planning period. A monetary value is assigned to the average waiting time in the objective function and added to capital investment and battery costs, however, the authors do not show how the cost of waiting is estimated in practice for a strategic planning horizon. In max-covering-based capacitated CSLP models, the objective is to build a given number of charging stations in a way to maximize EV flow coverage during a given planning period. Upchurch et al. [172] suggest that, in this case, peak hourly demand data would be appropriate for measuring station capacity, so that, the number of vehicles simultaneously recharging at a given station does not exceed the maximum station capacity, in the worst-case scenario. In other works, queuing theory is used in order to estimate the average waiting time based on the knowledge of average arrival rate and service rate per hour at each station. As an example, in [39], a spatial–temporal model provides information on location and time of charging demand to a location model and a capacity determination model. The capacity determination model sets the required number of chargers per station in a way to minimize total charger construction cost and EV waiting cost, both measured per hour. The total cost of a charger is converted to a cost per hour by considering its life-cycle. In another example, Mak et al. [128] use capacity constraints imposing that, with high probability, the inventory of batteries at each swapping station does not exceed the maximum allowed number, given that inventory is expressed as a function of the EV arrival rate per hour. Yang [199] and Xie et al. [192] also model station occupancy through chance constraints in a way that in each station, the probability of waiting time being higher than a certain time limit is less than a given threshold. The probability is calculated using queuing theory at each node assuming Poisson arrival and multiple servers (chargers) and knowing the arrival rate and the service rate per time unit at each node. Note that Xie et al. [192] calculate the arrival rate per hour as the ratio of the total EV flow charging at each node to the number of hours in the planning period. Another way to model capacity constraints in battery swapping stations is proposed by Sun et al. [164] who use a robust formulation guaranteeing that inventory of fully charged batteries in each station is sufficient to satisfy battery swapping demand in every time interval under the worst case scenario.

To sum up, in CS location problems, capacity constraints are rather aggregate over the planning horizon or represented in a way to guarantee sufficient capacity in the worst-case scenario (peak demand). Their role is to help determine the best size for located facilities but as discussed above, day-to-day adjustment of capacity is necessary to avoid congestion and long waiting times. For charging stations, it is not easy to transfer demand from one location to another. However, with new technologies, it would be possible to inform drivers in real time on the actual congestion in every charging facility and let them decide accordingly the best charging alternative. It would even be possible to help them decide with adequate EV routing algorithms taking into account facility congestion.

2.3.2. Deviation decisions

The deviation flow refueling location model (DFRLM) has been introduced by Kim and Kuby [97] as an extension of the FRLM of Kuby and Lim [105] with the idea that drivers would accept to deviate from their shortest path if the deviation remains within an acceptable range. Such assumption might allow for covering more EV flow with the same budget but requires to determine which path should be used on each OD trip, given that longer paths have less chance to be taken by EV drivers. Hosseini et al. [73] propose a capacitated deviation flow refueling location model (CDFRLM) formulated based on the arc covering approach of Capar et al. [22]. Their model involves more flexibility than Kim and Kuby's [97] model, as it allows for the use of more than one path on the same OD trip. The arc covering principle is also used by Yildiz et al. [203] to enhance their original path-segment-based formulation for the deviation-based CSLP.

Since the enumeration of all possible paths on every OD pair of a network is computationally demanding, most deviation-based models cited in Table 6 use, as input, a predetermined set of possible paths built in a way to keep a manageable size of the problem. A reasonable approach is to construct the set in a way that the amount of deviation from the shortest path on each OD pair does not exceed the maximum deviation tolerated by drivers. The k-shortest path algorithm with a deviation cap would then be implemented in a pre-processing step to provide the set of possible paths in single-level [73,77,97,98,114,115,192] or bi-level [54,147] CSLP models.

Only a few works (see Kınay et al. [99], Lin and Lin [120], Xu et al. [197], Yildiz et al. [203] and Zheng et al. [216]) explicitly incorporate routing decisions in the CSLP model in order to determine the path taken by each driver to travel from his origin to his destination without running out of charge. Yildiz et al. [203] develop a path-segment-based formulation with maximum deviation constraints and solve it using a branch-and-price algorithm. A similar path-segment-based formulation is considered by Kınay et al. [99] to model a full cover problem ensuring that drivers on every OD pair can complete their trip while minimizing the total cost of locating charging stations and the total en route recharging energy. The authors point out that since their model does not involve a predetermined deviation tolerance, it is more challenging to solve it to optimality on large-scale instances within reasonable computational times. Hence, they develop a Benders decomposition algorithm in which sub-problems are many-to-many shortest path problems and propose a new approach to build the dual solution for sub-problems in a way to accelerate the performance of the Benders algorithm by up to 900 times.

A compact mixed-integer programming formulation is presented by Xu et al. [197] to describe the charging logic and detour behavior, subject to a maximum detour constraint for every OD pair, while Lin and Lin [120] formulate a bi-level p-center flow-refueling location problem as a link-based non-linear integer program. The upper level determines the infrastructure design while the lower level involves vehicle range-constrained shortest path (VRCSP) sub-problems that determine the shortest path for each OD pair, constrained by the EV driving range and given that a vehicle may be refueled along its path when necessary. In order to solve the problem to optimality, the VRCSP algorithm is embedded in an implicit enumeration on binary location variables. Zheng et al. [216] model flow equilibrium in the lower level of their problem through a link-node representation using origin-based flows in the range constraints without path enumeration.

Furthermore, to overcome the difficulty of enumerating all possible paths within the CSLP model, heuristic methods and column generation procedures have been proposed in the literature. For instance, Kim and Kuby [98] develop greedy heuristic algorithms that dynamically generate at each step of the algorithm the shortest deviation path on each OD pair given the driving range and the charging stations selected in the previous step. Chen et al. [25], He et al. [61], Liu and Wang [125] and Wang et al. [180] apply a column generation approach incorporating a shortest path algorithm to solve a flow equilibrium problem on a restricted path set and gradually expand the path set in each iteration until the optimal solution is found.

2.3.3. Routing decisions

More complex decisions related to EV routing have also been considered in the CSLP literature. The reader is referred to the paper of Erdelic and Caric [40] for a survey on variants and solution approaches of EV routing problems. Typically, EV routing models consider that the demand of a given set of customers has to be delivered using EVs, starting their trip from a depot and ending it at the same depot. The objective is to determine EV routes including customer nodes and recharging nodes in a way that each EV can complete its trip without running of charge and that all demand is delivered to customers. In EV location-routing models, both the location of CSs and the routes taken by EVs are decided. Yang and Sun [201] is among the first papers to study this type of problem, considering vehicle routing and battery swapping location decisions in a same model that minimizes station installation costs and routing costs. Arias et al. [6] propose an extension of this work taking into account two additional terms in the objective function: penalization of infeasible routes due to insufficient battery autonomy and cost of energy losses on the power distribution network due to the installation of CSs. Li-Ying and Yuan-Bin [118] introduce time windows and vehicle capacity in the problem while Paz et al. [142] consider the multi-depot case with time windows. Furthermore, Schiffer and Walther [155] propose a location routing-model where different recharging options are allowed, including partial recharging, full recharging and recharging at customer sites. They also examine various minimization objective functions involving the total driven distance, the number of vehicles used, the number of CSs located as well as the investment and operational costs. In subsequent works, the authors consider the location of generic intra-route facilities that can be used by EVs to unload freight (customer nodes), to recharge the battery ([156,157]) or for both types of resources ([154]). A different application of EV location-routing is proposed by Zhou and Tan [219] in the context of an automotive assembly line, where a fleet of identical capacitated EVs is responsible of delivering material in small frequent lots from a central warehouse to assembly line stations (customers).

Table 7
Multi-period and stochastic CSLP models.

		Single-period	Multi-period	
		Stochastic	Deterministic	Stochastic
Deviation	Capacitated	[199,218]	[3,41,133]	[192,205]
	Uncapacitated	[112]	[115]	[179]
No deviation	Capacitated	[39,70,128,145,193,200]	[23,71,113,123,170,209]	[2,88,89,144,164,166,210]
	Uncapacitated	[36,84,94,95,111,151,189]	[27,33,35,139,152]	[32,34,90]

2.3.4. Technology choice

In view of the different EV recharging options available on the market and of their different costs and performances, technology selection seems to be one of the most important decisions that network planners have to make. Table 6 shows that several flow-based CSLP models integrate technology choice decisions together with the main decisions of CS location and demand coverage. For instance, Davidov and Pantoš [34] consider technology selection from a set of generic technologies having each a different recharging rate and an uncertain operation cost. Wang and Lin [182] analyze set covering and maximal covering approaches to determine the location of three types of CSs: battery swapping, slow charging or fast charging. The proposed models take into account different attributes for each station type with regard to cost, capacity and recharging rate. The same charging modes are taken into account by Wang et al. [179]. The authors first use the FISK's stochastic traffic assignment model [42] to determine the traffic flow on each OD pair. Based on the obtained flow distribution, they determine the best locations for slow charging, fast charging or battery swapping stations, given uncertain EV charging loads, with the objectives of minimizing investment costs, minimizing energy loss in the power distribution network and maximizing the annually covered traffic flow. Zhang et al. [212] and Liu and Wang [125] also consider path selection decisions together with technology choice decisions. Finally, the tri-level model proposed by Liu and Wang [125] involves technology selection in the second level while station locations are decided in the upper level and traveler's routing user equilibrium is determined in the lower level. Liu and Wang [125] distinguish between two types of wireless charging options: static option that requires parking to recharge and dynamic option that allows to recharge while driving.

2.4. Uncertainty and time-dependency

EV charging infrastructure planning and more generally facility location involve strategic decisions with an impact spanning a long time horizon. The parameters of a facility location problem such as demand and costs may vary widely during the planning horizon and thus become difficult to predict in an accurate way when the network design decisions are being made. For this reason, stochastic facility location problems have been studied by many researchers who have proposed different models and solution approaches to handle uncertainties (see the literature review on FLP under uncertainties by Snyder [161]). Furthermore, in long term planning, decision makers have to deal with parameters that change over time and need to adapt their decisions to these variations. In this case, a multi-period model offers the possibility of making decisions dynamically according to the evolution of the problem parameters (a survey on dynamic FLP is given by Arabani and Farahani [4]). Typically, in the context of EV charging infrastructure planning, multi-period modeling allows for opening the required number of CSs in every time period based on the recharging demand and the budget available instead of making a one-shot decision for the entire planning horizon. It also allows for adjusting more operational decisions such as vehicle flow and recharging plan according to the network conditions in each period. Articles considering multi-period and/or stochastic CSLP models are relatively recent and account for around 25% of the CSLP literature. Table 7 shows a classification of these articles according to multi-period and stochastic components as well as station capacity consideration and possibility for drivers to deviate from their shortest paths while Table 8 identifies the uncertain parameters in stochastic models.

2.4.1. Stochastic models

A first observation of the stochastic model survey in Table 7 shows that deviation and CS capacity features are omitted by many articles in this category. This simplification aims at reducing the number of constraints and variables in the problem and seems understandable in view of the complexity inherent to stochastic problems. Among the recent studies in this category, de Vries and Duijzer [36], Kchaou-Boujelben and Gicquel [94,95] and Lee and Han [111] propose new models and solution approaches to handle stochastic CSLPs under driving range uncertainty. In deterministic CSLP models, the battery SOC after recharging is assumed to be constant and fixed to the battery maximum capacity while the energy consumption is considered proportional to the distance traveled. This is why in this case, the driving range can be stated as a fixed distance (e.g. 200 km) calculated by dividing the battery capacity (e.g. 40 kWh) by the unit energy consumption (e.g. 0.2 kWh/km). However, in practice, the battery SOC after recharging is subject to uncertainties and depends among other on the time spent in the CS, the age of the battery and its technology. Moreover, the unit energy consumption is variable according to the road topology, the traffic, the driving style, the use of cooling or heating devices, etc. Therefore, the driving range is variable and its uncertainty should be taken into account in CSLP models in order to obtain more accurate location solutions.

de Vries and Duijzer [36] and Kchaou-Boujelben and Gicquel [94] use in their models a single random variable representing the driving range over the entire road network. They study two types of stochastic problems, the expected flow refueling location problem that evaluates the probability of coverage of each trip then maximizes the total expected EV flow covered and the chance

Table 8
Uncertain parameters in stochastic models.

	Demand	Service	Range	Trip	Cost or price	Energy supply	Power grid parameters	Travel/charging time
An [2]	✓							
Davidov and Pantoš [32]			✓					
Davidov and Pantoš [34]			✓	✓	✓			✓
de Vries and Duijzer [36]			✓					
Dong et al. [39]	✓	✓						
Hosseini and MirHassani [70]	✓							
Jing et al. [84]					✓			
Jung et al. [88]	✓							
Kabli et al. [89]	✓							
Kadri et al. [90]	✓							
Kchaou-Boujelben and Gicquel [94] [95]			✓					
Lee and Han [111]			✓					
Lee et al. [112]			✓					
Mak et al. [128]	✓							
Rajabi-Ghahnavieh and Sadeghi-Barzani [145]							✓	✓
Sathaye and Kelley [151]	✓							
Sun et al. [164]	✓				✓			
Tafakkori et al. [166]					✓			✓
Tao et al. [167]	✓						✓	
Wang et al. [179]							✓	
Wu and Sioshansi [189]	✓							
Xie et al. [192]	✓							
Xie et al. [193]	✓					✓		
Yang [199] [200]	✓	✓						
Yildiz et al. [205]	✓							
Zhang et al. [210]	✓							
Zhou et al. [218]	✓							

constrained flow refueling location problem that defines trip coverage using probabilistic constraints then maximizes the total EV flow covered. Kchaou-Boujelben and Gicquel [95] and Lee and Han [111] relax the assumption of having the same driving range over the entire network. Lee and Han [111] consider multiple random variables representing the driving range on each portion of the road but assume that these variables are independent from one another. Kchaou-Boujelben and Gicquel [95] propose a more detailed representation of the uncertain driving range using correlated random variables representing each the unit energy consumption on a portion of the road in addition to a single random variable for the battery SOC after recharging. One possible explanation of the correlation between energy consumption values on different portions of the road is that a change in the traffic intensity in a given area might influence the traffic in surrounding areas.

Another type of uncertainty that should be considered when planning EV charging infrastructure is the one related to customer demand. Predicting the pace of EV adoption by drivers and determining an accurate forecast for the amount and location of the future demand for EV recharging is indeed difficult (e.g. Hu et al. [74] propose to predict future demand using a Bayesian regularization neural network approach). Therefore, accounting for demand uncertainty in CSLP models would lead to a more realistic analysis and a better station location plan. Kadri et al. [90] evaluate the benefit to be around 9% of additional EV flow covered when implementing the solution of a stochastic model under uncertainty on the recharging demand instead of the deterministic solution based on average demand setting. As shown in Table 8, most works on stochastic CSLPs have focused on studying demand uncertainty. For instance, Sathaye and Kelley [151] represent uncertain recharging demand using density functions. They propose a continuous optimization approach to find optimal station density instead of determining precise locations on the road network. Wu and Sioshansi [189] formulate a two-stage stochastic program for the flow capturing location model under EV flow volume uncertainty, while Schiffer and Walther [157] and Zhang et al. [208] consider the uncertainty of the demand delivered to a set of customers by an EV logistics fleet. They apply robust optimization in solving a stochastic EV location routing problem when the uncertainties of customer demand and service time windows are modeled using a set-based representation.

Introducing additional features in the stochastic model such as station capacity restrictions and the possibility of deviating from the shortest path leads to more complex problems but enables the decision maker to carry out a more realistic analysis of the trade-offs relating costs, customer convenience and recharging demand satisfaction in an uncertain environment. For instance, Dong et al. [39] and Yang et al. [200] develop capacitated CSLP models under uncertainties on demand and service rates. They formulate service level constraints using queuing theory, assuming that arrival and service rates at CSs follow a Poisson process. Another example of capacitated stochastic problem is proposed by Mak et al. [128] who study the location of capacitated battery swapping stations under recharging demand uncertainty. Station capacity restrictions are expressed using chance constraints imposing that (with high probability) the number of batteries at a station does not exceed the maximum allowable number that can be simultaneously recharged by the electric grid.

With regard to the possibility of deviating from the shortest path, Table 7 suggests that there is very little work integrating this feature in stochastic and/or dynamic models. This might be explained by the computational difficulty of the resulting problems that should combine path selection decisions with uncertainties on the input parameters, which might lead to a large number of scenarios if uncertainty is discretized. Similarly, a large number of time-periods leads to computational difficulties in multi-period problems. The size of the problem typically increases with the number of possible paths on each trip, number of time periods in the planning horizon and number of scenarios in case random variables are discretized. Examples of stochastic models considering path selection decisions can yet be found in Lee et al. [112] and Yang [199]. Lee et al. [112] propose a bi-level optimization model where it is assumed that every trip q is shorter than the driving range but the EV is not necessarily fully charged at the origin O_q of the trip. The remaining fuel range at O_q is considered variable and thus might be insufficient to complete the trip, in which case the driver needs to stop once for recharging. Accordingly, the authors incorporate in their objective function the minimization of the travel failure rate, calculated on each trip q as the probability of having the remaining fuel range at O_q shorter than the distance to the closest charging station (or to D_q if it is closer). Yang [199] study a capacitated CSLP where a driver can make a detour from his shortest path to recharge his vehicle and can choose the CS to visit based on a maximum utility selection constraint. Uncertainty in this model concerns the recharging demand and service rates and is represented through a linear congestion constraint stating that the risk of having the waiting time at CSs higher than a given threshold should be low enough.

2.4.2. *Multi-period models*

One of the first multi-period extensions of the flow based CSLP was proposed by Chung and Kwon [27] who introduce time dependency of the EV flow volume in the path-segment-based formulation of MirHassani and Ebrazi [134] and determine optimal decisions for CS location in each time-period. Hosseini and MirHassani [71] study a capacitated dynamic CSLP based on the same path-segment-based formulation. They use the expanded network technique but introduce in their network time nodes and time links in addition to geographical nodes and geographical links. This allows them to evaluate traveling time in their objective function and formulate constraints on the maximum waiting time in queue. Davidov and Pantoš [33] [35] develop multi-period models based on a node covering approach, where a trip is considered covered if each node belonging to the trip can be refueled at every time period of the planning horizon by a CS situated within a distance shorter than the driving range. Furthermore, the arc covering approach of Capar et al. [22] is used by Zhang et al. [209] to develop a multi-period capacitated CSLP model. The authors represent the EV recharging demand growth over time using two components, the first one is a natural growth applicable to the entire network and the second one is specific to each path. The specific growth depends on current demand coverage resulting from the infrastructure built in the past. This dependency leads to non-linear components in the objective function and constraints, hence the problem solution is handled using heuristics. Furthermore, Fakhrmoosavi et al. [41] study a multi-period flow-based CSLP minimizing total investment costs and monetary value of detour and queuing delays. The model involves seasonality of traffic demand and battery performance and it is shown that, with regard to station locations, the reduced battery performance in cold weather is a more critical factor than the increased demand in warm seasons. Different modeling approaches have been adopted in other works, where Cavadas et al. [23] investigate a node-based capacitated multi-period CSLP that takes into account the effects of peak hours on the problem solution while Tu et al. [170] use actual GPS data of conventional taxis in Shenzhen, China, to build a spatial–temporal approach aiming at maximizing taxi electrification.

It is also important to note that, according to Table 7, the literature pertaining to multi-period stochastic CSLP remains limited. To the best of our knowledge, the only flow-based model using multi-stage stochastic integer programming was presented by Kadri et al. [90]. The authors propose to use a scenario tree to represent the evolution of uncertain recharging demand over time and develop exact and heuristic solution methods to solve the problem on instances of realistic size. Other multi-period stochastic CSLPs are studied by Davidov and Pantoš [32] [34] who extend the multi-period deterministic model of Davidov and Pantoš [33] to the stochastic context. Davidov and Pantoš [32] consider the uncertainty of the driving range only while Davidov and Pantoš [34] integrate uncertainties on the range, the trajectories taken by drivers, the disposable charging time and the costs related to each charging technology. Moreover, Jung et al. [88] study a stochastic dynamic itinerary interception problem for electric taxis under uncertainty on taxi demand. They propose a two-level iterative approach that seeks to minimize the average delay of taxis composed of the time to reach the charging facility, the waiting time before recharging and the service time. Zhang et al. [210] develop a two-stage stochastic program that determines location and capacity decisions on coupled transportation and power distribution networks, under dynamic and uncertain demand. The strategic decisions related to the station locations and to the number of charging spots per station are static and made in the first stage while the operational decisions related to the volume of EVs recharging at each station and power/voltage management decisions are dynamic and occur in the second stage when uncertain demand is revealed. Finally, Sun et al. [164] determine optimal battery swapping station locations, battery inventory level and recharging plan at each station under two types of uncertainties: EV flow volume uncertainty and electricity price uncertainty.

3. Solution methods and applications of the charging station location problem

We complete our review of the literature on charging station location by analyzing different approaches to solving the problem and to applying the proposed models on real-life data.

3.1. Solution methods

As with other facility location problems, solving the CSLP is challenging for large size instances involving a substantial number of variables and constraints in the model. In addition to binary location variables, flow-based CSLP models include binary OD pair coverage variables and eventually binary flow variables associated with refueling logic constraints used to model the recharging of EVs at different nodes on their way from their origin to their destination. Since the number of possible OD pairs, nodes on each OD pair and deviations from each shortest path in real-life networks are often very large, the problem can quickly become intractable. Other challenges in solving the problem may also arise if the objective function or constraints are non-linear and eventually non-convex, which is likely to occur when the model involves stochastic variables, a bi-level structure or integration of the electricity network with the transportation network. These situations generally require the development of approximation approaches to reach near-optimal solutions in reasonable computation times.

In this section, we survey the different solution approaches adopted in the literature to tackle CSLPs. Fig. 5 depicts the most frequently used solution methods. Note here that we focus our analysis on single objective models since this category accounts for the major part of the literature. In Fig. 5, the class "Solver" refers to the use of an off-the-shelf solver to solve the problem either to optimality or until a "good enough" solution is obtained or a given time limit is reached. The class "Approximate" involves the papers that propose an approximate reformulation of the studied problem because the original formulation is difficult to solve as is, mainly due to non-convex expressions. The problem is then usually solved using an off-the-shelf solver based on the approximate reformulation. Classes "Exact" and "Heuristic" both refer to the implementation of an adhoc solution method either to solve the problem to optimality ("Exact") or to find near-optimal solutions ("Heuristic"). As shown by Fig. 5, the most popular approaches used in solving CSLPs are heuristic methods (43% of the papers) and off-the-shelf solvers (38%). Typically, a solver can be employed to solve small-size instances of the problem within a reasonable amount of time (e.g. [132]). However, the instance size becomes significantly larger when one needs to deal with realistic situations, particularly with regard to the number of variables and constraints. This could explain the popular use of heuristics (e.g. [131,165]) to obtain good quality solutions within short computation times when this is not possible by simply running a solver. In the following, we review the CSLP literature considering heuristic, approximate or exact solution approaches.

3.1.1. Heuristic methods

Early attempts by Berman et al. [13] and Hodgson et al. [64] to solve the FCLM using greedy heuristics show that this type of approach leads to good quality solutions within short computation times. Lim and Kuby [119] employ greedy heuristics to overcome the large number of refueling combinations when applying the FRLM to the real-life case-study of Florida. Although their numerical tests show that greedy adding heuristics perform well in solving large-size instances of the problem, the authors conclude that a genetic algorithm (GA) would provide slightly better solutions. Greedy heuristics have also been implemented in the case of deviation-based CSLP under limited driving range ([114]) as well as stochastic capacitated ([70]) and multi-period capacitated flow-based CSLPs ([71] and [113]). Vazifeh et al. [174] propose to combine a greedy search with a genetic algorithm to solve a node-based set covering CSLP minimizing the excess driving distance to reach charging stations.

Genetic algorithms (GA) have been widely used in the CSLP literature [37,61,90,91,115,119,184,192,206]. Important features of GA have been considered in these works, such as chromosome representation, initial population setting, efficiency of fitness function evaluation as well as adequate mutation and crossover operators. For instance, to solve their multi-stage stochastic program, Kadri et al. [90] propose a genetic algorithm where the initial population of candidate stations uses a restricted set of "promising" vertices of the network instead of random selection of vertices. In addition, the fitness function evaluation is made more efficient by employing an arc-covering-based approach that is carried out only once in a pre-processing step. Li et al. [115] apply an enhanced version of the GA-based heuristic developed by Vose [177], including feasibility check and solution refining for their multi-period multi-path CSLP. An adaptation of the same GA approach is used by Xie et al. [192] to solve a stochastic capacitated extension of the latter problem. Wang et al. [184] develop a GA based on the FRLM of Kuby and Lim [105] and design each chromosome as a sequence of all refueling combinations. They compare the GA performance with the one of a problem-specific heuristic method and also study the combination of the two approaches. The combined heuristic method appears to be the most performant in efficiently solving the problem.

In addition to genetic algorithms and greedy methods, other heuristic approaches have been proposed in the literature such as simulated annealing metaheuristic (see e.g. Ghamami et al. [47], Lee et al. [112] and Zockaie et al. [221]), adaptive large neighborhood search (Guo et al. [54]), lagrangean-relaxation-based heuristic (Hosseini and MirHassani [72]), tabu search-based heuristic (Kchaou-Boujelben and Gicquel [94]), chemical reaction optimization metaheuristic (Lam et al. [110]), hybrid evolutionary algorithm (Tian et al. [168]), whale optimization algorithm (Zhang et al. [214]) and particle swarm optimization (Xu et al. [196]). In particular, combined heuristic procedures have been proposed to solve complex location-routing CSLPs including greedy methods with simulated annealing (Hong et al. [68]), variable neighborhood descent and scatter search (Li et al. [117]), adaptive variable neighborhood search with tabu search (Li-Ying and Yuan-Bin [118]) as well as adaptive large neighborhood search with local search and dynamic programming components (Schiffer and Walther [156]).

3.1.2. Approximate methods

Complexity of certain CSLPs arises from the introduction of non-linear components in the objective function or constraints of the problem, which may lead to non-convex problems. This is usually the case, for example, when taking into account randomness of the problem parameters, bi-level decision making structure or integrated modeling of power distribution network and transportation network. One way to overcome the resulting computational difficulties is to find good approximations of the corresponding optimization problems. For instance, Kchaou-Boujelben and Gicquel [95] study a stochastic CSLP under the uncertainty of battery energy status and electricity consumption. The initial integer programming formulation involves a series of joint chance constraints and is thus difficult to handle as is. The authors propose two approximate approaches, the first one is based on the approximation of each joint constraint using Bonferroni's inequality while the second one employs partial sample approximation (PSA) of the random variables i.e., applying Monte Carlo sampling of the continuous distribution of all the random variables but one (the stochastic battery energy status). Similarly, Wu and Sioshansi [189] develop an L-shaped sample average approximation decomposition to solve their two-stage stochastic flow capturing location problem under EV flow volume uncertainty while Quddus et al. [144] use a solution method combining sample average approximation with a progressive hedging algorithm for their two-stage stochastic MIP that integrates both strategic and operational decisions. Mak et al. [128] cope with non-linear terms in the objective function of their robust swapping station network design model, which uses distributionally robust optimization, by introducing an approximation of the problem based on mixed integer second-order cone programming. Distributionally robust optimization technique is also employed by Xie et al. [193] to determine the capacities of photovoltaic panels and storage units in each station of a charging network working with solar energy. The problem is handled through reformulations based on risk theory that lead to mixed integer linear programs.

Non-linearity in stochastic CSLP models may also result from modeling customer serviceability in charging stations using queuing theory. Yang et al. [200] cope with such non-linearity by applying exponential regression and logarithmic transformation to derive linear approximations of maximum reject rate constraints at charging stations while Zhang et al. [211] employ piecewise linearization to reformulate station service capacity constraints into mixed integer linear constraints. Bi-level models typically involve challenging non-linear components as well. In Riemann et al. [147], for example, the flow on every possible route is a variable to be solved in the lower level of the problem, which leads to non-linear flow coverage objective function. The problem also includes non-linearity in the model constraints emanating from the use of the multinomial logit (MNL) model stochastic user equilibrium (SUE) principle as well as non-linear travel time function on each path. Therefore, the authors develop a linear approximation of the model before solving it using a commercial solver. Other examples of approximation approaches are considered by Cui et al. [31], Wang et al. [183] and Zhang et al. [210] Zhang et al. [211] to handle non-linear power flow equations in CSLP models coupling power distribution network and transportation network. Cui et al. [31] also use linear approximation of piece-wise step functions representing protective device costs as well as an assumption on the number of charging spots in each station to linearize distribution system expansion costs.

3.1.3. Exact methods

A classical approach to solve CSLPs and more generally facility location problems is to use Branch & Bound algorithms (see e.g. He et al. [56] and Miralinaghi et al. [133]), in which cutting planes (Arslan et al. [9], Göpfert and Bock [51] and Yildiz et al. [205]) or column generation (Xu and Meng [195] and Yildiz et al. [203] [204]) can be integrated in order to handle the problem more efficiently. The Branch & Cut algorithm developed by Arslan et al. [9] uses a natural formulation for the refueling station location problem with routing, while the one proposed by Göpfert and Bock [51] is based on the new concepts of OD-covers and OD-cuts. Yildiz et al. [205] also develop a Branch & Cut algorithm for their stochastic capacitated set covering CSLP. They first devise a cut formulation by projecting out the flow allocation variables from the original compact formulation of the problem. Then, they generate cuts when violated inequalities are detected. A Branch & Price algorithm is employed by Yildiz et al. [203] for their path-segment-based deviation flow refueling location model, taking into account driver deviation tolerance. They consider a branching process for location variables only, where column generation is used to calculate an upper bound at each node of the Branch & Bound tree. The pricing problem can be simply solved by defining a set of plausible node pairs for which it is possible to travel without recharging and without violating driver deviation tolerance constraints. A similar approach is used by Yildiz et al. [204] to solve a path-segment-based problem related to network design with relay. A Branch & Price algorithm is applied to the problem formulated as a multi-commodity flow problem and strengthened by optimality cuts. The pricing problem considered by Xu and Meng [195] in their deviation-based problem is however more challenging to solve due to the consideration of non-linear elastic demand. The authors thus develop a customized two-phase approach to handle the pricing problem.

Another popular way to efficiently solve CSLPs to optimality is to use decomposition techniques, such as Benders decomposition (BD). As stated in Melo et al. [130], the presence of binary (strategic) location decision variables together with continuous (tactical) flow decision variables in discrete facility location problems makes them attractive candidates for the application of decomposition techniques, capable of reducing the computational effort by solving a series of sub-problems, generally easier to handle than the original problem. Arslan and Karasan [8] develop a Benders decomposition algorithm to solve the CSLP for plug-in hybrid electric vehicles based on the arc covering approach of Capar et al. [22]. They test three methods for cut generation, single cut, multi-cut and single Pareto-optimal cut then show in their numerical experiments that the latter method is the most efficient one. Arslan and Karasan's [8] Benders decomposition approach using Pareto-optimal cut generation is extended by Kadri et al. [90] to a multi-period stochastic context. The numerical tests carried out by the authors on randomly generated instances of different sizes show that the multi-cut approach outperforms the single-cut one. Kınay et al. [99] develop a Benders decomposition algorithm where location decisions are determined in the master problem and optimal paths between OD pairs are constructed in the sub-problems. Solving

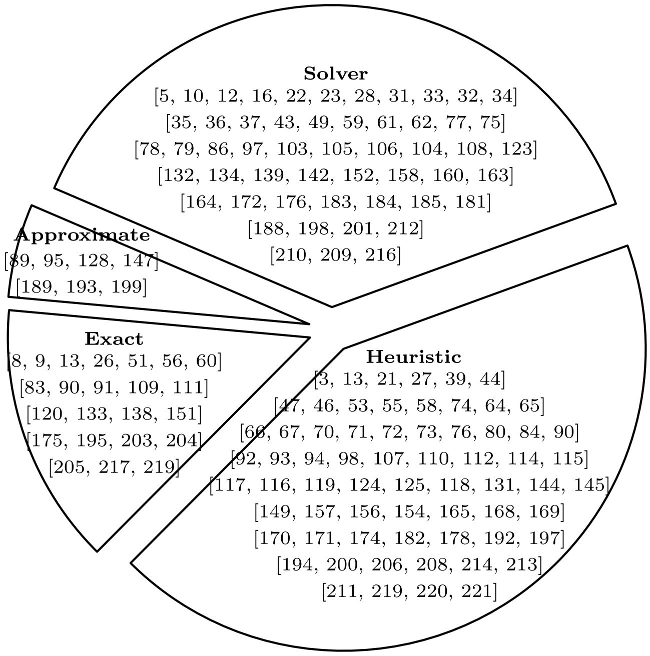

Fig. 5. Classification of CSLP papers according to solution approaches.

sub-problems requires around 95% of the overall computational effort, thus, the authors develop a novel solution methodology. When the sub-problem is feasible, the optimal dual solution is built to generate non-dominated optimality cuts; otherwise, valid inequalities are determined based on the infeasible sub-graph structures and added to the master problem as feasibility cuts.

Other applications of Benders decomposition to solve the CSLP to optimality are presented by Zheng and Peeta [217] and Lee and Han [111] while An [2] and Nourbakhsh and Ouyang [138] use decomposition techniques based on Lagrangian relaxation (LR). Nourbakhsh and Ouyang [138] apply LR on the constraints linking location variables and scheduling ones then decompose the main problem into two types of sub-problems: fixed-charge facility location and production lot-sizing, where the former is simply solved by inspection and the latter is solved to optimality after reformulating it into a network shortest path sub-problem. An [2] study a stochastic charging station location and fleet sizing problem for electric bus systems with dynamic charging plan. The author applies LR on two sets of constraints: those linking station location and bus assignment decisions and those stating capacity restrictions in charging stations. Similarly to Nourbakhsh and Ouyang [138], decomposition here leads to location sub-problems that can be solved by inspection and to charging sub-problems that are more challenging to solve.

Another notable challenge in several recent CSLP studies is to handle bi-level modeling structures. For instance, Chen et al. [25] and He et al. [60] consider bi-level CSLPs where in the lower level, drivers determine their path and battery charging plan according to travel time and other preferences, given charging infrastructure built in the upper level. Taking into account the resulting equilibrium conditions in the CSLP leads to formulating a mathematical program with complementarity constraints, that is difficult to solve for large instances. Chen et al. [25] and He et al. [60] thus use an active-set algorithm solving first a restricted version of the problem upon a pair of active sets corresponding to a particular deployment plan. Then, the active-sets are adjusted by solving a binary knapsack problem constructed based on the obtained Lagrangian multipliers and a better deployment plan can be derived. The procedure is iteratively repeated until there is no better plan.

3.1.4. Comparison of solution methods

As explained above, various solution methods have been proposed in the literature to tackle CSLPs. Therefore, it may be useful to show here some comparison of these methods in order to understand the trade-off between solution time and quality. Fair comparison of solution methods requires to apply them on the same instances with the same assumptions and on a same computing machine. Since the number of tests, the assumptions on problem parameters (such as vehicle range or number of stations to be built) and the computer used to run the program may differ from one article to another, direct comparison of the methods seems very difficult. Therefore, we only intend here to present an overview of the average performance of some exact and heuristic methods, used on the same test instances. We consider large-size benchmark instances that have been employed in more than one article, namely, the California network instances used by Yildiz et al. [203] composed of 339 nodes, 617 arcs and 1167 OD pairs and the Florida network instances introduced in Kuby et al. [107] consisting of 302 nodes, 495 arcs and 2701 OD pairs. Tables 9 and 10 illustrate the average performance of a set of methods on California and Florida instances respectively. The first column represents the reference in which the numerical results were given, the second column describes the type of computer used by this reference to run the

Table 9
Comparison of solution methods for flow-based CSLPs on California instances [203]
Size: 339 nodes, 617 arcs, 1167 OD pairs.

Ref.	Computer	Approach	Method	CPU(s)	Gap(%) (Time limit)
Arslan and Karasan [8]	3.16 GHz 2xDuo CPU 2GB RAM	Arc covering, hybrid EVs [8]	Solver BD Pareto-Optimal-Cut	874 57	0 0
Kchaou-Boujelben and Gicquel [94]	2.5 GHz Intel i5-3210M 8 GB RAM	Path-segment, explicit range [36] Path-segment, explicit range [94]	Solver Solver	28498 1430	0.5 (10 h) 0
Tran et al. [169]	2.8 GHz Intel i7-4810MQ 16 GB RAM	Node covering [21] Path-segment [134] Arc covering [22] Arc covering [169]	Solver Solver Solver Relaxation-based heuristic	394 301 114 5	0 0 0 0
Yildiz et al. [203]	AMD Opteron(tm)Processor 6282 SE 2 GB RAM	Path-segment, deviation [203]	Branch&Price	4772	0.02 (3 h)
Xu and Meng [195]	3.4 GHz Intel (R) Core (TM) Duo	FRLM, deviation, elastic demand [195]	Branch&Price	7243	0.1 (5 h)

Table 10
Comparison of solution methods for flow-based CSLPs on Florida instances [107]
Size: 302 nodes, 495 arcs, 2701 OD pairs.

Ref.	Computer	Approach	Method	CPU(s)	Gap(%) (Time limit)
Kchaou-Boujelben and Gicquel [94]	2.50 GHz Intel i5-3210M 8 GB RAM	Path-segment, explicit range [36] Path-segment, explicit range [94]	Solver Solver	21260 149	1.3 (10 h)
Capar and Kuby [21]	1.6 GHz Intel Core Q-720 6 GB RAM	Heuristic [119]	GA Greedy heuristic Greedy w/substitution	2745 21 254	0.26 0.4 0.29
Tran et al. [169]	2.8 GHz Intel i7-4810MQ 16 GB RAM	Node covering [21] Path-segment [134] Arc covering [22] Arc covering [169]	Solver Solver Solver Relaxation-based heuristic	401 202 56 51	0 0 0 0

tests, the third and fourth columns provide information on the model and the solution method, while the last two columns show the average CPU time and percentage optimality gap, calculated over all tests on California or Florida instances used in the paper.

With regard to flow-based problem formulation, the comparison provided in Tran et al. [169] suggests that the problem can be the most efficiently solved by a commercial solver when the arc covering modeling approach of Capar et al. [22] is used (average CPU is 114s in California case and 56s in Florida case). The path-segment-based approach of MirHassani and Ebrazi [134] requires longer running times (average CPU is 301s in California case and 202s in Florida case) while the node covering method shows the worst computational performance (average CPU is 394s in California case and 401s in Florida case). It is also notable that the relaxation-based heuristic proposed by Tran et al. [169] allows to solve the problem to optimality on Florida and California instances in shorter computation times (average of 5s in California case and 51s in Florida case). Other heuristic methods (GA, greedy, and greedy with substitution) are proposed by Lim and Kuby [119] and compared by Capar and Kuby [21] on Florida instances. Table 10 shows that GA provides on average the best solution (lowest gap) among the three heuristics but requires much longer computation times than the greedy heuristics.

Extensions of the arc covering and path-segment modeling approaches have been proposed by different authors to deal with realistic situations such as uncertain demand or vehicle range, use of hybrid EVs, possibility of deviation from shortest paths and station capacity constraints. For instance, Kchaou-Boujelben and Gicquel [94] and de Vries and Duijzer [36] propose path-segment-based formulations with the driving range explicitly included in the model, in order to introduce range uncertainty. The formulation presented by Kchaou-Boujelben and Gicquel [94] shows significantly better efficiency than the one proposed by de Vries and Duijzer [36] as optimal solutions are obtained in all instances within much shorter computation times. Arslan and Karasan [8] apply Benders decomposition (BD) based on Pareto-Optimal cut to solve a CSLP with plug-in hybrid electric vehicles on California instances. This

Table 11
Comparison of heuristic solution methods for battery swap station EV LRP on benchmark instances from Yang and Sun [201].

Instances	Ref.	Method	CPU(s)	Gap (%)
P-Instances: 16–101 customers	Schiffer et al. [154]	AVNS [66]	11.7	0.42
		ALNS [154]	34.6	0.57
	Schiffer and Walther [156]	AVNS [50]	87.3	0.26
		ALNS [156]	37.1	0.16
T-Instances: 50, 75, 100 customers	Schiffer et al. [154]	AVNS [66]	29.6	0.57
		ALNS [154]	68.8	0.84
	Schiffer and Walther [156]	AVNS [50]	206.9	0.06
		ALNS [156]	87.1	0.97
GW-Instances: up to 480 customers	Schiffer et al. [154]	AVNS [66]	640	2.8
		ALNS [154]	1698	3.6
	Schiffer and Walther [156]	AVNS [50]	1624	0
		ALNS [156]	1978	4.5

leads to solving the problem to optimality faster than a commercial solver (57s vs. 874s), however, BD method for this model has not been compared to other heuristic or exact methods. For more complex models, it would be even impossible sometimes to generate solver solution as a basis of comparison with other methods. For instance, Yildiz et al. [203] explain that they could not solve their deviation-based CSLP using a solver due to the exponential growth in the number of possible paths when the deviation tolerance level increases. For that reason, Table 9 only shows the average performance of their Branch & Price algorithm that allows to obtain very good solutions (0.02% optimality gap on average) within three hours of calculation. Similarly, Xu and Meng's Branch & Price algorithm [195] leads to near-optimal solutions to their deviation-based FRLM with elastic demand within five hours of calculation.

Combined CS location-routing problems (LRPs) are also very challenging to solve due to the high number of possible route combinations. Compared to classical LRPs, additional constraints are used in the model to ensure charging when required and completing the trip from the origin to the destination without running out of charge. That is why, heuristics are generally used in this case. Based on the two recent works of Schiffer et al. [154] and Schiffer and Walther [156]), we compare in Table 11 four solution methods: the adaptive variable neighborhood search (AVNS) algorithms developed by Goeke et al. [50] and Hof et al. [66] and the adaptive large neighborhood search (ALNS) algorithms proposed in Schiffer et al. [154] and Schiffer and Walther [156]. Note that direct comparison of methods is possible here as the same computer is used to run the tests in Schiffer et al. [154] and Schiffer and Walther [156] and the tests are done on the same instances (instances P, T and GW defined by Yang and Sun [201]). The results show that the AVNS methods perform better overall. The AVNS method proposed by Hof et al. [66] yields the highest performance with regard to CPU time but produces solutions that are of medium quality compared to the other methods, while the AVNS method developed by Goeke et al. [50] obtains the best quality solutions for medium (T) and large size (GW) instances but are not the fastest in solving the problem. This situation shows the usual trade-off between solution quality and running time faced when solving facility location problems, hence the selection of the best solution method depends on the priority set by the decision maker.

3.2. Applications

In the last decade, there has been several attempts to apply CSLP models in solving real-life problems of refueling infrastructure design for alternative fuel vehicles. As shown by Table 12, most case-studies used in flow-based models have been carried out in the USA. Another major part of applications deals with networks in East Asia (China, Taiwan, South Korea) and Europe. This can be explained by the fact that these regions of the world are among the most advanced ones in adopting EVs and other types of vehicles that use alternative fuel such as hydrogen and natural gas. However, for a case-study to be effective and yield accurate results that help in decision making, a thorough data analysis is necessary to fine-tune the model, in particular for the road network structure and the refueling demand estimation. Although real-life data may be retrieved from traffic information made available by some governments or communities (Chung and Kwon [27]), surveys (Dong et al. [37]) and specialized firms such as taxi (Yildiz et al. [205]) or bus (Xylia et al. [198]) companies, obtaining all the required information on future vehicle flow and refueling demand remains a challenging task. Therefore, researchers often resort to assumptions in order to complete their case-studies. For instance, as illustrated in Table 12, the gravity model can be used to estimate the flow volume on road trips when these data are missing (see e.g. Kuby et al. [107], Lee and Han [111], Upchurch et al. [172] and Yildiz et al. [203]). Other studies present a more detailed adhoc approach to estimate future AFV flow volume. Wu and Sioshansi [189] consider the case of the 12-county Central-Ohio region that covers around 6000 km^2 and use tour-record data of 1.1 million light-duty vehicles provided by the Mid-Ohio Regional Planning Commission. By assuming a 3% EV penetration rate, they randomly select from the tour-record data vehicle flows that should be electrified. This leads to a set of OD pairs and to the value of EV flow volume on each pair, for which the authors calculate shortest-time paths based on average vehicle speed available in the dataset. Note that in some models, estimating vehicle flow volume on every trip is not required (cases referred to as "NA" in Table 12). For instance, Li and Huang [114] propose a set covering model for the South Carolina road network. The objective is to minimize station investment costs while ensuring that at least one path, either a shortest or deviation path, is available between each OD pair in the network, which allows all drivers to

Table 12
Overview of case-studies used in flow-based CSLP models.
(NA: Not Applicable; NM: Not Mentioned)..

Case	Year	Country	Demand estimate	# Nodes	# *OD* Pairs	Vehicle type	Charging	Citations
Florida (intercity) [107]	2009	USA	Gravity model with demand weight	302	2701	Hydrogen	Fast	[21,22,36,67, 94] [95,98, 119,169,171]
Arizona [172]	2009	USA	Gravity model	50	300	AFV	Fast	[70]
Taiwan [181]	2009	Taiwan	NA	51	1100	Electric	Fast	–
Alexandria (Virginia) [160]	2011	USA	Adhoc Estimate	2476	114942	AFV	Fast	–
Penghu Island [182]	2013	Taiwan	Adhoc Estimate	25	12	Electric	Mixed	–
South Carolina [114]	2014	USA	NA	519	210	AFV	Fast	[115]
Denmark [188]	2014	Denmark	Adhoc Estimate	4209	2269	Electric	Fast	–
Korean Expressway Network [27]	2015	South Korea	Adhoc Estimate	324	5162	Electric	Fast	[28]
Pennsylvania Turnpike System [78]	2015	USA	Adhoc Estimate	38	171	Gas	Fast	[79,176]
California [203]	2016	USA	Gravity model	339	1167	AFV	Fast	[8,94,95,169]
Hainan Island [39]	2016	China	Adhoc Estimate	47	1081	Electric	Fast	–
Mashhad [221]	2016	Iran	NM	1429	15437	AFV	Fast	–
German Highway Network [86]	2016	Germany	Adhoc Estimate	595	5451	Electric	Fast	–
Mashhad [133]	2017	Iran	Adhoc Estimate	935	7157	Hydrogen	Fast	–
Central Ohio [189]	2017	USA	Adhoc Estimate	1805	1628110	Electric	Fast	–
LNG refueling infrastructure in EU [104]	2017	Europe	Adhoc Estimate	6812	9712	Gas	Fast	–
Texas Highway Network [111]	2017	USA	Gravity model	640	435	Electric	Fast	–
Stockholm Bus Network [198]	2017	Sweden	NA	403	143	Electric	Fast	–
California Natural gas infrastructure [153]	2017	USA	Adhoc Estimate	NM	NM	Gas	Fast	–
Hubei District [54]	2018	China	NM	57	110	Electric	Battery Swap	–
Hebei Highway Network [184]	2018	China	Adhoc Estimate	174	1653	Electric	Fast	–
California [192]	2018	USA	Adhoc Estimate	532	141246	Electric	Fast	–
Jeju Island [199]	2018	South Korea	Adhoc Estimate	82	1681	Electric	Fast	–
Yangtze River Delta Network [180]	2019	China	Adhoc Estimate	140	36	Electric	Fast	–
US Interstate Highway Network [56]	2019	USA	Adhoc Estimate	846	38220	Electric	Fast	–
Eifel Rural Region [152]	2019	Germany	Adhoc Estimate	110	44	Electric	Fast	–
Chicago [205]	2019	USA	Adhoc Estimate	192	88935	Electric	Fast	–
Greater Hartford area [215]	2019	USA	Adhoc Estimate	806	264196	Hydrogen	Fast	–

(*continued on next page*)

Table 12 (*continued*).

complete their journey without running out of charge. However, this would involve a much higher investment as all trips must be covered, as opposed to maximal covering approaches that maximize the flow volume covered within a given budget.

Another challenge of applying CSLP models on real-life cases relates to the size of the optimization problem. In practice, as indicated in Table 12, road networks might involve hundreds/thousands of nodes and thousands/millions of possible OD pairs. Among the cases surveyed in Table 12, the Central Ohio case seems to be the largest one with regard to the number of OD trips (1628110) while the Liquefied Natural Gas (LNG) refueling infrastructure in Europe involves the highest number of nodes of the network (6812). The large size of these real-life instances obviously leads to significant computational difficulties as well as extensive running times. Shukla et al. [160] show in their numerical tests that computation times for solving the CSLP on the Alexandria network (2476 nodes and 114942 trips) may exceed 100000 sec. Kadri et al. [90] analyze the impact of the network size and number of trips on computational performance. They point out that an increase in these numbers negatively impacts the performance of solution methods including Cplex solver, Benders decomposition and genetic algorithm. They also note that an increase in the number of trips appears to have higher impact on the performance. This could explain the wide use of adhoc exact and heuristic solution methods for real-life large-size instances as discussed in Section 3.1. Techniques to reduce the instance size have also been adopted in many case-studies. For instance, Chung and Kwon [27] use the traffic volume information provided by Korea Expressway Corporation in 2011 to construct a road network involving 104652 OD trips. To reduce the instance size, they consider in their study a subset of trips having vehicle flow volume higher than a certain threshold (for volume higher than 20000, the number of trips is reduced to 5162). Jochem et al. [86] apply their model to the case of the German Highway network that initially contains 19600 OD pairs. To generate a test instance of a manageable size, the authors remove the pairs with a distance lower than 40 km as drivers on the corresponding trips do not need recharging on their way. They also remove OD pairs involving a flow volume lower than 5000 vehicles per year. This leads to reducing the number of pairs to 5451 with only 7% decrease in the total flow volume considered. Another example can be found in He et al. [56] who use a clustering approach to diminish the size of the OD nodes set. This allows them to limit the scope of the case-study to a 196x196 OD matrix (38220 pairs) instead of an initial 4486x4486 OD matrix (more than 25 million pairs).

Even if the CSLP has been primarily studied for the planning of EV or AFV refueling infrastructure, applications in other contexts have been considered by researchers. For instance, Ribeiro et al. [146] investigate the use of unmanned aerial vehicles for carrying out periodic inspections of the belt conveyors employed in the mining industry. They develop a location-routing model to optimize CS locations and vehicle routes in a way to minimize the total system costs.

4. Discussion and perspectives for future research

4.1. Discussion

The problem of optimizing charging station locations is a complex problem that involves different decision levels, constraints and objectives. While early works on CSLP have tended to simplify the problem modeling, recent works have tried to incorporate more realistic features as explained in previous sections. However, more effort needs to be made in future works to enhance the practical relevancy of the models, particularly integrating features that have been so far studied separately. Extensions of existing works can be built along different lines including the followings: charging capacity impact, coupling of transportation and electricity networks, integration of novel charging technologies and measurement of environmental impact.

As discussed in our literature review, capacity constraints and waiting time issues have been addressed in many recent CSLP papers. Since demand is expected to rapidly grow in the future, charging capacity will become a more pressing issue and further investigation of the impact of capacity constraints on the problem decisions and performance should be considered, both from the investor and user perspectives. For instance, Table 3 shows that only a few works have integrated station capacity constraints in a bi-level model. Examples include Chen et al. [26] and Huang and Kockelman [76] who propose bi-level modeling approaches to optimize both location and capacity of charging stations where the lower level problem models charging and route choice equilibrium minimizing individual driver travel and waiting time, given location and capacity decisions made in the upper level. In practice, EV drivers take into account charging facility congestion and waiting time when selecting the combination of facilities that will refuel their route, hence, it is important to integrate station capacity limitation in bi-level models as one of the important decision drivers at both upper (investor) and lower (user) levels. Furthermore, according to Table 6, developing CSLP models that combine station capacity decisions with other types of decisions such as the possibility of deviation from shortest paths, routing and technology choice appears as a potential perspective for future research. While the focus of early works on a limited set of decisions is understandable, future works should investigate the interrelation between different decisions and examine the resulting trade-offs when planning the charging infrastructure.

Another complexity of the problem stems from the interdependence of the transportation network and the power grid. EV drivers would select their routes based on availability of charging station capacity while effective operation of the power grid strongly depends on recharging demand realization at every time period. Several recent works have realized the importance of integrating the two types of network in CSLP models. He et al. [58] point out that integration is important to capture the influence of EV recharging load on the power grid when making the decision of CS location. Kabli et al. [89] combine power grid expansion and charging station location decisions. They develop a multi-period two-stage stochastic program where first stage decisions involve power grid expansion while second stage decisions concern locations of new charging stations in areas with sufficient power supply resulting from prior power grid expansion. Another example is given by Lin et al. [123] who suggest that, due to the high power requirement of electric buses and their extensive charging time, it is important to couple the transportation network planning with the optimization of the power supply system. They study a multi-period CSLP combining the siting and sizing of charging stations with an optimal power flow problem. The recent literature on CSLP thus shows a clear trend involving power grid and transportation network coupling. However, in view of the increasing complexity of the problem in this case, obtaining models that are more realistic, technically applicable and tractable at the same time would require a combined effort from operations research and electrical engineering research communities.

Moreover, it is important that CSLP models consider the impact of introducing new technologies on charging infrastructure planning. For instance, wireless charging is among the new charging technologies that allow to recharge the battery without plugging the vehicle into a source of energy. Liu and Wang [125] highlight four major benefits of using dynamic wireless charging: recharging while driving which results in shorter charging time, saving the space that should have been allocated to CSs, removing the risk of electric shock and reducing the battery pack weight, hence increasing the vehicle speed. Recently, Ngo et al. [136] and Iliopoulou and Kepaptsoglou [80] have proposed bi-level programming approaches to locate dynamic wireless charging facilities. They use continuous variables to model facility location decisions as opposed to binary variables used in past studies, which allows to select a proportion of a road link instead of the entire length of the link to build a wireless facility. In future studies, it is particularly important to examine the trade-off between investment costs and quality of service as well as match the right charging technology with the customer needs. For instance, wireless charging lanes would require that drivers spend a certain time driving so that to have a sufficient amount of energy recharged, which might oblige the driver to reduce his speed, hence delay his arrival to destination.

Finally, it is notable that little attention has been paid to environmental objectives in CSLP models while the ultimate objective of promoting the use of EVs instead of conventional ICE vehicles is the reduction of greenhouse gas emissions. EVs are not direct source of pollution but one should not ignore the indirect pollution in this case [135]. This includes emissions from the sources of energy that refuel the EV charging infrastructure as well as the management of end-of-life batteries. In future works, investigating the environmental impact of EVs might thus prove important. Comparing EVs with conventional vehicles or other alternative fuel vehicles such as hydrogen-based ones with regard to the economic, service and environmental aspects is necessary to determine the right power generation mix and accordingly build the required infrastructure. As example, Tafakkori et al. [166] and Tao et al. [167] propose to measure the environmental impact of the transportation network, including direct pollution resulting from using conventional gasoline vehicles and indirect pollution generated by the refueling structure of AFVs.

4.2. Challenge of application and parameter estimation

One of the major challenges to overcome when applying CSLP models to real-life studies is the estimation of input parameters such as recharging demand and energy consumption. A detailed demand forecasting including the recharging amount, its time of occurrence and its location would allow for a better planning of the charging infrastructure and a more appropriate matching of energy supply with recharging demand. However, the production of accurate demand forecasts capturing spatial and temporal distribution of recharging requests is currently hindered by the lack of data since the scope of EV use is still limited.

Though data on conventional ICE vehicle trips can be available for case-studies as discussed in Section 3.2, mainly from GPS records of taxi fleets (see e.g. Cai et al. [20], Li et al. [116], Shahraki et al. [158] and Yang et al. [200]), freight companies (Bai et al. [11] and Hwang et al. [78]) or volunteer households (Dong et al. [37]), the representativeness of such data for EVs needs to be further investigated as argued by Cai et al. [20]. Not only EV drivers may have different travel patterns than ICE vehicle drivers but also it is difficult to capture the recharging behavior of EV drivers by analyzing ICE vehicle travels. Li et al. [116] and Tu et al. [170] attempt to simulate drivers' travel and recharging behavior based on large conventional taxi trip dataset and assumptions on taxi recharging conditions such as a minimum dwell time of the taxi and a predefined threshold for battery SOC. However, as in most existing literature, it is assumed in these works that energy consumption is proportional to the distance traveled, hence ignoring the impact of road configuration, uphill vs. downhill travel, weather constraints, traffic conditions and driving style on the vehicle consumption. This assumption could be misleading and result in less accurate forecast of recharging requests. It is thus important to consider relaxing this strong assumption in future research work in order to model energy consumption in a more realistic way.

In one of the recent works, Kchaou-Boujelben and Gicquel [95] develop a stochastic CSLP model under uncertain energy consumption on each portion of the road, so as to incorporate the impact of driving conditions on the state of charge of the battery. However, the authors assume energy consumption to follow a given probability distribution and do not discuss how to estimate it in practice. In addition to travel surveys and GPS records of ICE vehicles, small scale EV trials can be used as a source of data. A recent work by Crozier et al. [30] combines travel survey information with high resolution data from an EV trial in order to build a stochastic data-driven model that estimates recharging demand taking into account variability of travel behavior and recharging behavior i.e., circumstances under which the driver decides to recharge. With the increasing number of EVs on the road and the deployment of smart chargers, larger scale EV data including trip information as well as time and location of recharging requests will become more easily available, which will facilitate the development of more advanced forecasting models using big data and artificial intelligence capabilities (see e.g. Al-Ogaili et al. [1] and Arias and Bae [5]). Feeding CSLP models with more detailed and accurate forecasts would pave the way for future development of appropriate charging infrastructure.

4.3. Outlook on future research directions

The next decade is likely to see conventional ICE vehicles banned in many countries as strategic plans have been developed to reach 100% zero-emission vehicles on the road. Norway is one of the most advanced countries in this field with plug-in passenger vehicles representing 70.2% of new vehicles purchased in August 2020 [207]. Other countries such as the UK, Sweden, the Netherland, Canada and Spain are expected to follow with a target of achieving 100% zero-emission vehicles on the road between 2030 and 2040 [87].

To reach this goal, the actual barriers towards wide EV adoption should be mitigated in the near future. Technological advances in vehicle and battery development would lead to producing more affordable EVs with longer driving range. In addition, the large investments planned by governments to expand their charging infrastructure would result in a more convenient charging experience for customers and will alleviate the range anxiety of many users. However, future challenges would rather emanate from the increasing demand for recharging and the need to recharge large capacity batteries in a short time. In the future, the capacity problem will not only concern the increasing amount of demand but also its temporal and spatial distribution. Namely, during peak hours, a large number of recharging requests may arise at the same time and location, which would result in a high load on the power grid and consequently satisfying all demand would be more challenging. To adapt the grid to support charging load peaks, substantial investment will be required for high-voltage and expensive transmission circuits as well as new transmission substations, switches and service transformers [150]. To mitigate charging peaks and reduce infrastructure and energy costs, decision makers need to first determine an accurate demand forecast capturing the variation of the recharging amount, time and location, as discussed in the previous subsection. Then, they should identify optimal solutions to match energy supply with recharging demand, in particular when the use of renewable energy sources such as solar and wind leads to variable energy supply.

One possible solution to overcome supply and demand variability is to employ energy storage solutions (ESS) in order to smooth electricity production requirements. As suggested by Gallinaro [45], ESS could serve for energy storage when supply is higher than demand, typically during off-peak demand periods then the stored energy could be used to satisfy recharging demand during peak hours. Another solution consists in influencing the recharging demand pattern through variable pricing. The idea here is to design charging rates in a way to dissuade customers from charging at times and locations where the demand is high or where the supply is low. Shaping demand would result in better matching of energy supply with recharging demand at lower costs. According to simulations made by Boston Consulting Group [150], an optimized charging behavior would minimize the investments required to upgrade the grid and would allow for a reduction in transmission and distribution costs per EV of about 70% by 2030 compared to the non-optimized case. Examples of relevant works recently published include the adaptive pricing model of Valoggianni et al. [173] and the charging station sizing model of Khaksari et al. [96] integrating scheduling algorithms for smart recharging.

In view of these new challenges, future research should adopt a more comprehensive approach in modeling charging infrastructure planning. Given an accurate forecast of the amount, time and location of recharging demand, CSLP models should determine optimal location and capacity of charging stations together with optimal solutions to match supply and demand including investments to adapt the grid where needed, location and sizing of energy storage solutions, selection of appropriate supply sources as well as scheduling and pricing of EV recharging. Integrating demand shaping strategies in CSLPs may call for the development of new models to capture the interrelation of charging rates and demand. Moreover, new approaches would be required to elaborate realistic and tractable CSLP models that combine long term strategic decisions with more operational decisions such as scheduling and pricing. A recent example of such approach is given by Quddus et al. [144] who develop an integrated two-stage stochastic program where first-stage decisions relate to the location and capacity of stations (charging or battery swapping) while the second-stage decisions pertain to battery storage in swapping stations, hourly usage of available power sources and battery charging/discharging. Obviously, dealing with more comprehensive models in a dynamic and stochastic context would result in new challenges for solving complex mathematical problems and would thus necessitate innovative solution methods to tackle computational difficulties. Particularly, since our review showed that most papers used commercial solvers or heuristic methods as solution approaches (Fig. 5), there should be more focus on developing exact methods that provide optimal solutions within acceptable computation times.

5. Conclusions

In this paper, we have provided a comprehensive review of modeling features, solution methods and applications of the charging station location problem (CSLP). We have investigated optimization objectives, decision variables, and constraints taken into account by researchers as well as game theoretical perspectives, particularly bi-level models. We have also discussed recharging demand representation techniques and distinguished between two widely used approaches, namely node-based and flow-based representations. Accordingly, we have analyzed different ways of modeling demand coverage in flow-based models and outlined typical mathematical formulations in each case. Moreover, as a strategic planning problem, CSLP is characterized by its stochastic and dynamic nature, thus requiring the integration of parameter uncertainty and dynamic decision making in the planning process. We have thus reviewed stochastic and multi-period models proposed in the literature and discussed the types of uncertainties considered. We have completed our review by exploring various solution methods and applications of the problem and concluded by discussing some open issues and new trends, which has allowed us to identify future research directions in the field.

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